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Toward health promotion and prevention: Evidence from a food and health partnership model of care

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Abstract

Health promotion and disease prevention requires health systems address the patients' social needs using new care delivery models. Yet, research in this area has stalled for several reasons. We study a partnership model of care that couples clinical care delivered by primary care providers and social services delivered by community-based organizations, and its impact on patients' preventive health outcomes and behaviors. We use data from the Mid-Ohio Farmacy, which is a collaboration across the Mid-Ohio Food Collective (MOFC), a network of 650+ affiliated food pantries, and a large federally qualified health center (FQHC). The FQHC offers primary and preventative healthcare services across eight free clinics, which are co-located with the MOFC-affiliated food pantries. Patients were screened for food insecurity during their clinic visit and, if positive, were referred to the Farmacy. Compliers made at least one visit to the food pantry after referral, while noncompliers did not. Using difference-in-differences, we find that compliers had no discernible change in their body mass index (BMI, kg/m²), which we refer to as a BMI stabilization effect. Non-compliers' BMI increased after referral. High comorbid and high pantry use compliers experienced a significant reduction in their BMI and a marginally significant reduction in glycated hemoglobin (HbA1c, %). These patients had unique compliance behaviors, including greater search, frequency, and consistency of food pantry use. Travel costs suggests that high comorbid patients ascribed a greater value to the Farmacy program. In terms of primary care utilization, we find that compliers' clinic visit patterns after referral were consistent with the visit patterns observed in the food secure cohort, suggesting that the Farmacy program may have helped compliers address competing demands that are known to inhibit health behaviors.

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KEYWORDS

community-based organization, Farmacy, health promotion, partnership model of care

Highlights

- The food and health partnership model of care helps stabilize patient health outcomes, curbing the progressive risk of chronic conditions like obesity, hypertension, and diabetes.
- Patients with co-morbid conditions had greater search, frequency, and consistency of food pantry use after referral, suggesting that these patients were more likely to use “food as medicine.”
- In terms of primary care utilization, food insecure compliers' clinic visit patterns after referral were consistent with the visit patterns observed in the food secure cohort, suggesting that the partnership model helps compliers address competing demands that inhibit health behaviors.

1 | INTRODUCTION

Healthcare systems and their associated clinical processes—which focus on the diagnosis, treatment and management of acute health conditions in the short term—are ill-suited for health promotion and disease prevention over the long term. The global shift toward preventative and population-based models of care has prompted many open-ended calls for a new models of care delivery (Deloitte, 2019). The proposed ideas accentuate a broader healthcare ecosystem to include new entities (Dai & Tayur, 2020) and patient routing patterns (Betcheva et al., 2020; KC et al., 2020), with the explicit intent on “keeping the population healthy for longer” (Keskinocak & Savva, 2020, p. 69). One concept that is gaining traction in the operations management literature is external collaboration (Adepoju et al., 2023).

External collaboration in primary care requires health systems work closely with new, non-traditional health entities that provide social services, intended to address the patient's social determinants of health (SDOH).¹ Connecting patients to non-profit community based organizations (CBOs) acknowledges the fact that the patient is not exogenous to the healthcare delivery process (Roels & Staats, 2021). Instead, the patient is a co-producer (Bendoly & Oliva, 2024) and shares the responsibility for their health outcome (Johnston, 2019). When a healthcare system partners with an external organization to address patient health, we refer to this approach as a partnership model of care.

Integrating CBOs into healthcare delivery creates several opportunities for health promotion. The main advantage is that CBOs may be better suited at ensuring health equity, meaning that everyone has a fair opportunity to

attain their full health potential (Hill-Briggs et al., 2021). If it is easier for a patient to access the CBO, relative to a healthcare clinic, then CBOs may also be better suited to advise patients on routine health-related behaviors and use low-cost diagnostic exercises such as diet recalls to inform health behaviors. These exercises are particularly important for low-income patients, who have notoriously poor healthcare access. Patients with chronic conditions are also likely to benefit from more routine care, as they must attend to interrelated, complex and multi-dimensional components of care over the long term. If CBOs can monitor transient changes in the patients' social needs, to include important effect mediators like nutrition security (Mozaffarian et al., 2021; Seligman et al., 2023), then partnership models could also help prevent health complications and reduce the need for high intensity clinical interventions. Therefore, partnerships with external CBOs may benefit small primary care practices in a manner similar to the technologically-intensive patient-centered medical home, which has been adopted by large medical centers with the prerequisite infrastructure (Thompson et al., 2019).

Despite such benefits, however, partnership models of care can also lead to several operational challenges that are not well understood. For example, it is hard for health systems to evaluate the effectiveness of partnership models due to the lack of information exchange between health systems and CBOs. Determining program effectiveness requires the patients' protected health information is examined relative to their food consumption patterns, which can be difficult and expensive (Clark et al., 2020). Moreover, linkage rates or the rate at which a patient complies with a food referral and makes at least one visit to the food pantry can be very low, which casts

greater doubt on their effectiveness. Given these operational challenges and the lack of empirical evidence, partnership models have not yet scaled to solve long-term health issues. The purpose of this research is to assess the operational effectiveness and scalability of these models. By doing so, we extend the literature on care coordination (Dobrzykowski & Tarafdar, 2015; Kohnke et al., 2017) and examine the role of non-traditional health entities like CBOs in healthcare delivery.

The partnership model that we examine in this study can also be classified as a clinic-based food referral (DePuccio et al., 2022) or partnership-based food prescription program (Ranjit et al., 2023). It targets food insecurity as the principle SDOH with the idea that healthy food prescriptions can act as a therapeutic agent to improve health outcomes, aligning with the popular adage that “food is medicine” (Park, 2019). This model uses a direct referral, meaning that the clinic shares limited patient information directly with the partnering CBOs (Escobar et al., 2021). Direct information exchange closes the loop between the healthcare clinic, where the referral originates, and the CBO, where the referral is redeemed. In contrast, information-only or indirect referrals are more passive and essentially limit the resources offered to the patient. These referrals can take the form of a conversation, brochure or pamphlet that is provided at the clinic (Helms, 2019).

Barriers to partnership models of care, including perceived stigma, neighborhood deprivation or operational inaccessibility, have received considerable attention in the literature (De Marchis et al., 2019; DePuccio et al., 2022; Lowrey et al., 2023; Marpadga et al., 2019; Taylor et al., 2016). Ranjit et al. (2023) studied a partnership-based regional food prescription program and found that 60% of referred patients did not visit at least one of 1500+ available food pantries, and 85% made 12 or fewer pantry visits. Another study found linkage rates between the healthcare clinic and CBO as low as 0%–4% (Marpadga et al., 2019). Low linkage rates or program compliance pose significant data limitations (discussed in greater detail in Appendix Section A.1), which has stalled research in this area² (Swartz, 2018; Zenk et al., 2022). Our research seeks to address the above limitations by investigating two main research questions. The first research question is, does the Farmacy program improve health quality outcomes? Our main measures of health quality include body mass index (BMI, kg/m²) and glycated hemoglobin (HbA1c, %). Healthy BMI and HbA1c are well-recognized measures of health quality and important for disease prevention (DPP, 2002; Ward et al., 2019). Second, does the Farmacy program affect health behaviors? Clinic utilization serve as a proxy for health behaviors. We address these questions using

large-scale empirical data from the Mid-Ohio Farmacy program, which is a partnership model of care that links the Mid-Ohio Food Collective (MOFC) and affiliated food pantries to a large federally qualified health center (FQHC) in Franklin County, Ohio.³

The Mid-Ohio Farmacy program is an off-site program that uses a direct referral. A low-income patient that screens positive for food insecurity at the FQHC and complies with the referral by making at least one visit to the food pantry. Compliance results in greater access to healthy food. All data was collected by the MOFC. The third author (employed by the MOFC) directly managed protected patient health information and matched clinic health outcome data—available after the patient presented at any one of the eight FQHC clinics—with food pantry utilization data—available when the same patient visited any of the 330+ MOFC-affiliated food pantries—from June 2015 to May 2018. The patient population included adults diagnosed with a chronic metabolic condition, including obesity, hypertension, or diabetes. 80% of patients were at or below the federal poverty level (FPL = gross income at or below \$30,000 for a family of four), and 99% earned less than double that or below 200% FPL. 22,930 patients were diagnosed with a chronic metabolic condition, yet only 16% ($n = 3709$) screened positive for food insecurity and were referred to the Farmacy program.⁴ In the 1-year period after referral, 37% ($n = 1382$) made one or more visits to the food pantry, and 7.9% ($n = 294$) made 11 or more. Patients varied in a number of ways. For example, the timing of the patients' clinic and food pantry visits, relative to referral, complicates evaluation. We relied on strict inclusion criteria (Appendix Section A.1) and multiple levels of analysis (Appendix Section A.2) to address such limitations.

We estimate treatment effects with reference to three main cohorts. *Nonparticipants* or control patients screened negative for food insecurity (i.e., were food secure) and were not referred to the Farmacy. *Participants* or treated patients screened positive and were referred to the Farmacy. Participants had the option to comply with referral. A participant *complier* made at least one visit to the food pantry in the 1-year period after referral or had 1+ redemptions (Ranjit et al., 2023); a participant *noncomplier* had 0 redemptions. All patients are low-income and face similar struggles to manage their long-term health across the clinical and social spectrum. Using difference-in-differences, we find that noncompliers' BMI (kg/m²) increased after referral, which confirms the negative health consequence of food insecurity on health outcomes. Yet, compliers had no discernible change in their BMI after referral. This BMI stabilization effect suggests that the Farmacy program helped reduce the progressive risk of a metabolic disease like obesity.

We also found evidence of a potential dose–response relationship, whereby patients that were high comorbid at the time of referral (Charlson Comorbidity Index greater than 3) and achieved high pantry use compliers after referral (≥ 11 food pantry visits) experienced a significant reduction in their BMI and a marginally significant reduction in their HbA1c. To better understand these results, we examined multiple indicators of food pantry use among high pantry use compliers. We find that high comorbid and high pantry use compliers demonstrated different compliance behaviors and engaged with the Pharmacy program in a manner consistent with “food as medicine.” High comorbid patients may have ascribed a greater value to the food pantry service (Byrne & Just, 2023) or developed more creative solutions to make the visit, such as piggybacking pantry visits off of other planned trips (see Appendix Section A.9.1).

With respect to clinic utilization, the volume of clinic visits generally decreased throughout the week for both food secure and food insecure patients. However, food insecure noncompliers were significantly more likely to visit the clinic on Friday and Saturday, compared to food secure nonparticipants. The day-of-week coefficients for the food insecure complier group reveals interesting trends before and after referral. Before referral, we observed no significant intraweek variation in clinic utilization (i.e., the volume of clinic visits did not differ significantly between Saturday and Monday). However, after referral, clinic visit volume was significantly lower on Saturdays. A lower probability of a visit later in the week is consistent with the clinic utilization observed in the food secure nonparticipant group. This suggests that the Pharmacy program may have helped food insecure compliers address competing demands known to inhibit health behaviors such as scheduling and attending primary care appointments. These patterns are consistent with food secure patients, who typically make primary care visits early in the week.

The rest of the paper is organized as follows. Section 2 provides an overview of the related literature. Our hypotheses are presented in Section 3, and our data and empirical approach is detailed in Section 4. Results are reviewed in Sections 5 and 6. Section 7 concludes.

2 | LITERATURE REVIEW

The partnership between healthcare clinics and CBOs distinguishes partnership models of care from other patient or provider-centric care delivery programs aimed at coordinating activities across multiple locations. Clinic-community integration increases the scope of services offered and enables healthcare organizations to

address patients' health-related needs (Bonney & Chang, 2018; Feinberg et al., 2018). One of the key objectives of the partnership model is to address the care fragmentation issue, which is particularly challenging for patients with low socio-economic status (SES), by connecting patients to CBO services important for self-care. In this section, we review the HOM literature on how care fragmentation has been studied. This allows us to identify the gaps that our research seeks to address.

2.1 | Fragmentation in healthcare delivery and the role of the partnership model

Fragmentation of care is one of the greatest challenges facing healthcare systems in the US (Hwang & Christensen, 2008). Temporal, clinical, and geographical fragmentation of care pose a number of challenges that are difficult to manage within a single care delivery model (KC et al., 2020). In traditional healthcare models, patients are responsible for the coordination of their care that may require them to integrate instructions from their primary care doctors, specialists and other auxiliary services. The reason for the limited success of these healthcare models is due to the myopic focus on acute care delivery inside the hospitals' four walls (Rittenhouse et al., 2009). In the empirical HOM literature, many studies fail to consider healthcare as a people-centric operation (Roels & Staats, 2021). More specifically, these studies do not account for the patients' health behaviors following discharge or between medical appointments. These behaviors are known to impact health outcomes and clinic utilization via the individuals' willingness and ability to engage in their own care (Hibbard & Greene, 2013). Health-related social needs deserve more attention and explicit consideration in healthcare delivery.

In recent years, there have been some advances to such fragmented care through healthcare reforms such as the Accountable Care Organization (Lan et al., 2022) and Patient-Centered Medical Homes (Thompson et al., 2019). Related HOM studies examine coordination of care among primary care and specialty providers, but do not look into the coordination between non-clinical and clinical aspects of care. The partnership model of care that we study is typified by the Accountable Health Communities model, introduced most recently by the Centers for Medicare & Medicaid Services (CMS, gov, 2023).

Figure 1 outlines the existing solutions to healthcare fragmentation. These solutions focus on who and where healthcare services are delivered. Improved care

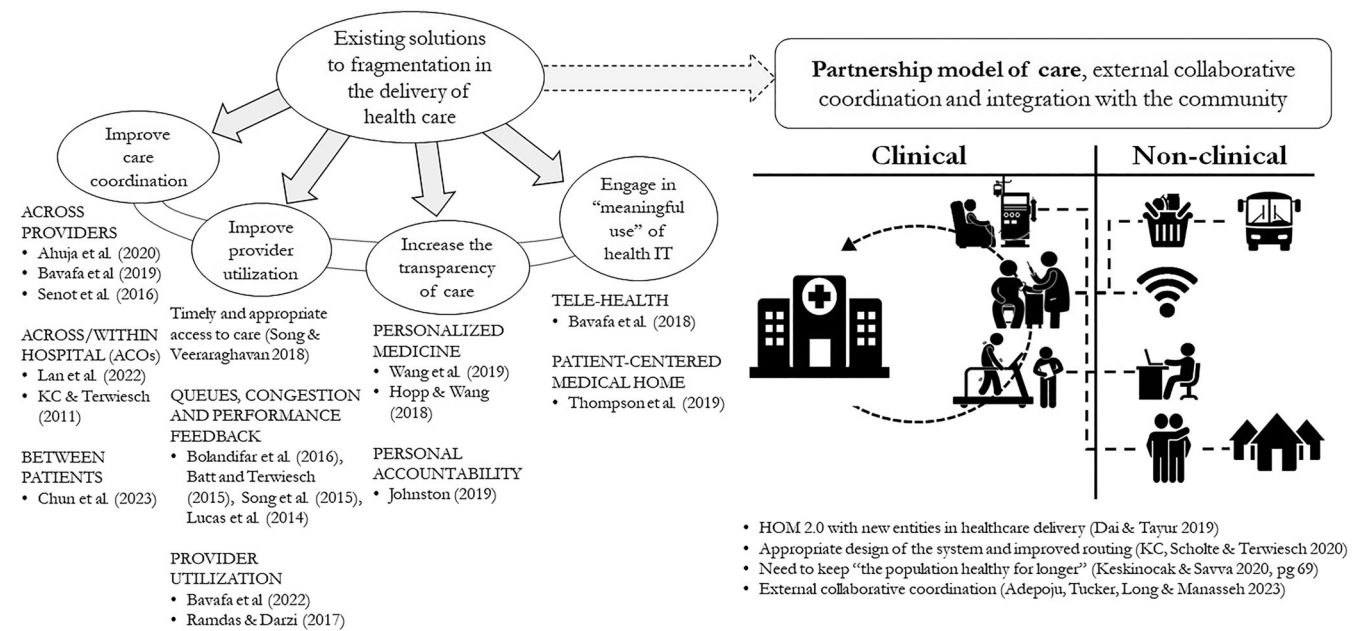


FIGURE 1 Existing solutions to healthcare fragmentation from the HOM literature.

coordination has mostly focused on the activities that occur between providers and within the hospital. For instance, Senot et al., 2016 find that improving communication practices between physicians and nurses can help better manage patients during their hospital stays. Kim et al. (2015) studies capacity management in a hospital setting and explores how admission control policies for an upstream unit impact operations downstream. A second stream of research has focused on improving provider utilization so that they are able to see more patients. In this domain, many studies frame healthcare as a queuing system and consider the patient as an “exogenous entity” (Roels & Staats, 2021). These studies primarily examine variation in process flows (Batt & Terwiesch, 2015), wait times (Lucas et al., 2014) and congestion (Bolandifar et al., 2016), in high-cost care channels such as the emergency room. In terms of operational transparency, Wang et al. (2019) and Hopp et al. (2018) find that greater information sharing enables better healthcare decisions—for both patients and providers.

Finally, a more recent line of research in information systems explores the “meaningful use” of health information technologies (HIT). It includes Thompson et al. (2019), who explore how EHR data can be used in a Patient-Centered Medical Home to prevent or displace high-cost complications for diabetics. Their case-control study was, in part, informed by similar limitations in preventative care measures. Another example is Bavafa et al. (2018), who characterizes the effects of the introduction of the e-visit channel on patient visit patterns, health, and provider availability. A revolutionary idea—albeit one that is still largely limited to the use of hospital

resources—is shared medical appointments (Ramdas & Darzi, 2017).

Our review of the literature reveals very few empirical studies on external collaboration, with the exception of Adepoju et al. (2023). External collaboration can take on multiple forms.⁵ Much like the fact that a patient–patient interaction could substitute for a patient–provider interaction during a shared medical appointment, it is also possible for volunteer–patient interaction at the CBO to act as a substitute (or complement). The volunteer–patient interaction has a similar capacity to improve patient health or health behaviors by shifting what is shared during the conversation. CBO volunteers are also likely to have greater awareness of community resources and can better guide the patients’ access to social services such as food. Providers that relegate this aspect of care to CBO volunteers as part of their external collaboration efforts could improve their care delivery efficiency. It could also enhance the social-relevance of treatment, which can improve relationships (Hassan et al., 2020) and patient satisfaction (Dobrzykowski & Tarafdar, 2015). In the following section we derive two broad sets of hypotheses related to the impact of partnership models of care.

3 | HYPOTHESES DEVELOPMENT

3.1 | The partnership model of care and preventative health quality

Our review of the literature suggests that partnership models of care could improve health outcomes in a

number of ways. Recent research has shown notable differences in health behaviors and food preferences based on SES, whereby those who are more affluent and live in areas with greater access to fresh foods may develop an inherent preference for those healthy foods (Allcott et al., 2019). There are many other factors, like stress, which have also been linked to SES with a partial effect on health outcomes (Baum et al., 1999). Adler and Newman (2002) find that chronically higher stress, a non-clinical factor of health, results in greater morbidity or mortality. Non-clinical factors have multiple dimensions that can complicate the design and intended effects of healthy eating interventions, especially price- or access-related ones (Levi et al., 2019).

Seligman and Schillinger (2010) were among the first to articulate the interplay between clinical and non-clinical factors of health in the cycle of food insecurity and chronic disease, followed by others (Garcia et al., 2018). Their cycle reveals several mechanisms by which the individuals' food security status impacts health. For example, food insecure individuals experience constrained options and compensatory behaviors that are conditional on their food environment. An individual may adopt different food consumption strategies depending on the availability of food, including overconsumption in periods of food adequacy and skipped meals in periods of food shortage. These compensatory behaviors contribute to volatile physiological changes, including the level of circulating blood sugars and rate of fat metabolism. The body's physiological response increases the individual's susceptibility to disease. For example, in periods of food shortage, the body may store more glycogen in the form of belly fat. This is a type of fat that secretes unhealthy hormones and makes disease more likely (Brody, 2018). Breaking the cycle by adjusting or eliminating these compensatory behaviors is difficult for patients who are resource-limited, especially if their food preferences and dietary patterns have significant cultural or social roots.

For these patients, subjective preferences and economic costs are two primary determinants of their enacted health behaviors. For example, the cost of a balanced diet (e.g., consumption of fruits and vegetables) typically competes with the cost of medical care: a healthy food choice may postpone the timing of a needed medication, while paying for a medication could lead to skipped meals. Berkowitz et al. (2014) refers to this routine as the "treat or eat" trade-off and find that cost-related medication underuse increases by a factor of four for those with social needs. To the best of our knowledge, the existing research in HOM has largely overlooked the more nuanced role of the patients' SDOH and the "plausibly causal" (Berkowitz, Basu, et al., 2018, p. 1612)

connection between food insecurity and health outcomes. Simchi-Levi (2020) has formally called for more research on health equity for marginalized populations, especially considering the inequitable aftermath of the COVID-19 pandemic. Our study offers a start.

The partnership model of care studied in our research identifies and addresses food insecurity, a SDOH attributed to an extra \$1863 in preventable health care expenditures per person per year, adding up to \$77.5 billion in annual healthcare costs in the United States (Berkowitz, Parashuram, et al., 2018; Morse, 2019). The higher cost of care associated with food insecurity may in part be due to compensatory behaviors that exacerbate chronic illness by increasing stress and reducing patient activation (Hibbard & Greene, 2013). To the extent that food insecure individuals are less willing or able to manage their own health, preventative health behaviors like diet or lifestyle changes may be impractical. These patients are not too different to those in global healthcare contexts who receive healthcare services in "resource-limited settings" (Jónasson et al., 2022). The lack of resources and lower patient activation for those who are food insecure reduces the aptitude for health promotion.

We argue that the partnership model of care, which connects food insecure patients to a community resource, helps to improve preventive health and reduce the risk of health-related social needs. This should be especially true for patients diagnosed with a chronic metabolic condition like obesity, hypertension or diabetes, which are partly diet related. In a partnership model, the patient is first identified as food insecure and then referred to a nearby food pantry. From the perspective of the care continuum, patient interactions with food pantry volunteers serve to supplement the patient-provider interactions that already occur at the healthcare clinic. However, patient-volunteer interactions are different in scope and purpose to patient-provider ones. In many instances, the volunteer may offer social support to the individual through intimate conversations about their community or familiar circumstances, in addition to the direct provision of social services like food. Since the location and hours of operation of a food pantry are usually more convenient than those of a primary care clinic, these interactions are usually more accessible, relatable and personable. Food pantry volunteers are also more likely to reside in the same community as the patient and better able to advise on community resources that promote and sustain healthy behaviors. On the other hand, healthcare providers may be unaware of such resources. Volunteer-patient interactions that occur outside the purview of the healthcare clinic serve to effectively outsource non-clinical components of care to the CBO partner. They complement clinical efforts to increase patient activation and health

promotion. In a small sample study conducted at the University of Chicago, patients who received information on community food resources reported higher levels of activation (Helms, 2019) and better health behaviors.

Relative to a healthcare clinic, CBOs like food pantries are much more accessible in terms of the numbers of locations, availability of staff, hours of operation, cost, and so forth. Better access generally translates to better outcomes, particularly in the healthcare setting where insurance and other financial barriers can greatly restrict access (Keskinocak & Savva, 2020). CBOs facilitate access by delivering social services for free or at reduced cost and in a convenient manner. For food insecure patients, a food pantry close to home or work may enable more reliable and consistent access to the healthy food required for a more balanced diet. In the Farmacy program evaluated in our study, a patient who visited the food pantry received a basket of food—including milk, eggs, fresh produce and meat—accounting for almost 2 meals a day for 2 weeks when adjusted for the patient's family size. When a patient (or any other client) checks in at the food pantry, a volunteer is assigned and accompanies the patient side-by-side during their pantry visit. During these “shop-throughs,” volunteers softly educate the client on how to prepare, cook, and store food items, recognizing that it may be the first time that the client has held or inspected the specific food. This means that patients not only get access to fresh foods, but also receive instructions and education on how to integrate the food into their dietary routine. The idea that a trip to the food pantry as a part of the Farmacy program could serve to increase the consumption of fresh food and, in turn, improve health, acts as the basis of our first hypothesis:

Hypothesis 1. Patients with chronic metabolic diseases (like obesity, hypertension, and diabetes) who participate in a partnership model of care will have better preventative health quality. The provision of non-clinical services addresses health-related social needs and lessens their negative effect on clinical health outcomes.

3.2 | The partnership model of care and clinic utilization

If partnership models of care help patients to address the competing demands posed by clinical and social needs—and, for example, enable the patient to eat a balanced diet and afford proper medical care—then we might expect a change in clinic utilization. A change in clinic visit

patterns should correspond with a change in the patients' aptitude for self-care and health promotion behaviors, given the strict resource-limitations faced by low-income patient cohorts (Jónasson et al., 2022). Interventions aimed at addressing health-related social needs are commonly evaluated based on changes in the demand for clinical care as a proxy measure for unobserved health behaviors.

Consider a food insecure adult patient who works a full-time job between the hours of 8 a.m. and 5 p.m. This patient is resource-limited, especially in the amount of time available to make a visit to their primary care doctor. Due to the demands of their full-time job, this patient may spend their limited resources (e.g., time or income) to ensure consistent access to food and postpone needed medical care received during scheduled doctors' appointments. If this were the case, the more urgent need for food would supersede a preventative or seemingly elective primary care encounter. It is also reasonable to assume that on days when the patient has fewer occupational demands—perhaps, later in the week on days like Friday or Saturday—they would be more willing to attend their doctor's appointments. Examining the timing of the primary care visit in a manner similar to Anderson et al. (2014) allows us to make an inference on the extent to which social needs have influenced clinical health behaviors and demand for care. We examine intra-week variation as our primary measure of clinic utilization.

Social needs affect the demand for care in several ways. If food insecure patients postpone needed care or miss appointments altogether, this temporal displacement of care could lead to worse health outcomes (Thompson et al., 2019) or a greater use of high-cost hospital resources like the emergency room (Berkowitz, Terranova, et al., 2018). Improper utilization of healthcare services may be partly responsible for an increase in the total cost of care (Berkowitz, Seligman, et al., 2018). Also, food insecure adults who receive electronic benefit transfers through federal feeding programs typically receive resources on a recurring cycle that has been linked to predictable patterns in health care usage. For example, adults who receive benefits from the Supplemental Nutrition Assistance Program (SNAP, formerly known as food stamps) tend to receive their benefits at the beginning of the month or in the first ten days. At the end of the month, when the monthly SNAP allocation has likely been exhausted, there is higher risk of hypoglycemic events and hospitalization for patients with Type 2 diabetes (Seligman et al., 2014). Exhaustion of SNAP benefits also leads to increased demand for other social services programs. Byrne and Just (2021) coined the idea of an “inverse SNAP cycle” whereby the number of food pantry visits markedly increases at the end of the month.

This is likely to be due to the exhaustion of household food budgets. Even when individuals lose access to SNAP benefits and experience a short-term disenrollment, the time burden required to reapply can disrupt healthcare utilization patterns (Xie et al., 2024).

If food insecure patients respond to competing demands by prioritizing food purchases over needed medical care, and also postpone the timing of primary care encounters due to their occupational demands, and food secure patients do not, then it is reasonable to expect differences in clinic utilization by food security status. HOM scholars have argued that social factors such as food insecurity can play an important role in affecting patient health outcomes (Keskinocak & Savva, 2020). Since social factors affect clinical health outcomes (Hussein & Collins, 2016), then these factors should similarly affect the demand for healthcare services. If food secure adults, who are not subject to competing demands to the same degree, have fewer social factors to contend with when scheduling and maintaining their preventative health appointments, then we might expect them to have more balanced healthcare visit patterns throughout the week. If participation in the partnership model of care helps food insecure patients address competing demands and minimize the trade-off between food and health (Berkowitz et al., 2014), then there might be a change in clinic utilization before and after referral. Ideally, the food insecure clinic visit patterns after referral would resemble those of food secure individuals. Therefore, we hypothesize:

Hypothesis 2. Patients with chronic metabolic diseases (like obesity, hypertension, and diabetes) who participate in a partnership model of care are better able to resolve competing demands. This is reflected by a change in clinic utilization before and after referral in a manner that is consistent with the clinic utilization patterns observed among food secure patients.

4 | DATA AND EMPIRICAL APPROACH

Given the challenges of cross-sector data sharing (Walker et al., 2022), we worked extensively with the lead dietitian at the FQHC to formalize a procedure for linking pantry and clinic data. Pantry utilization data was retrieved using *PantryTrak* (<https://secure.pantrytrak.com>), a proprietary client-relationship management tool developed by the MOFC and deployed across all food pantries to collect service encounter data. To resolve

concerns related to protected health information, we developed a geocode hashing program to convert each patient's physical home address to a randomized (± 500 -meter radius) geocoordinate location. The study protocol was approved by The Ohio State University Institutional Review Board.

4.1 | Setting

The first phase of the Farmacy program was a small-scale pilot that included 50 patients and three food pantries. The second-phase implementation, which linked the eight FQHC clinic locations to the larger network of 650+ MOFC-affiliated food pantries, provided the data for the present study. The main study area encompassed a territory of approximately 500 square miles with a total population of almost 1 million (MN) people. In 2023, the MOFC distributed nearly 80 MN pounds of food, 67% of it fresh (MOFC Annual Report, 2022).

The FQHC operates like a primary care clinic for uninsured or underinsured patients. The mid-level providers who mainly staff FQHC clinics offer routine and follow-up care, but not emergency services. Because low insurance coverage can pose a barrier to access, FQHC patient visits may be sporadic, and patients tend to travel to the clinics geographically closest to them. The FQHC receives federal funds and enhanced payments to provide care for uninsured and underinsured patients in its territory. 80% of patients were at or below the federal poverty level (FPL = gross income at or below \$30,000 for a family of four), and 99% earned less than double that or below 200% FPL. 30% of patients were best treated in a language other than English. State-level qualification criteria for food bank services, namely household-adjusted income $\leq 185\%$ FPL, and pantry-specific food distribution criteria, namely an individual must reside in a specific county, were modified such that all patients referred to this program qualified for food pantry services.

Flyers prominently displayed in FQHC waiting rooms across the eight clinic locations afforded all patients the option of discussing the Farmacy program with any primary care provider. Screening for food insecurity employed a validated two-item questionnaire referred to as the "Hunger Vital Signs" (Hager et al., 2010):

1. Within the past 12 months, we worried whether our food would run out before we got money to buy more.
2. Within the past 12 months, the food we bought just didn't last and we didn't have money to get more.

Patients who answered either or both questions with "sometimes true" or "often true" screened positive for

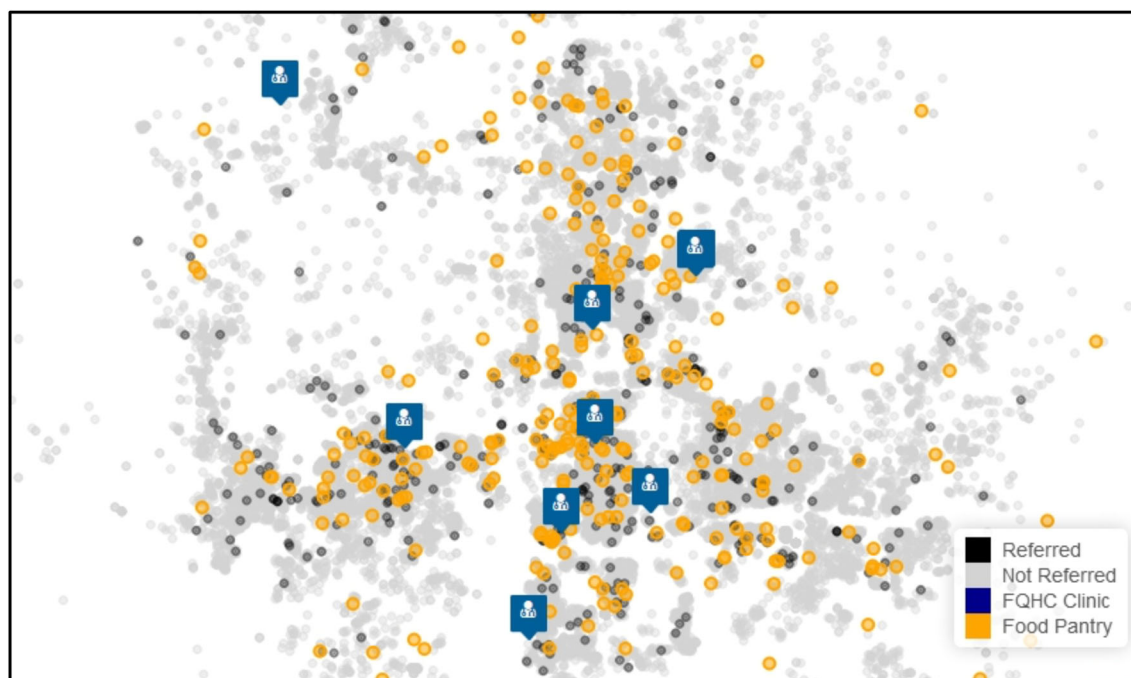


FIGURE 2 The study area includes the randomized home address of the 22,930 FQHC patient population with chronic metabolic conditions, eight clinic locations, and 330 (of 650+) MOFC member food pantries.

food insecurity and were referred to the Pharmacy program. Note that this threshold for referral was low. A patient that responded “never true” to both questions would be considered food secure and not receive a referral.

4.2 | Data

The data includes all patients diagnosed with a chronic metabolic condition who visited any one of the eight FQHC clinics between June 24, 2015 and May 11, 2018. The patient population included 22,930 unique individuals; 19,221 patients were screened as food secure and not referred to the Pharmacy program, and the remaining 3709 screened as food insecure. The incidence rate of 16.2% is close to other estimates for the study area.⁶ Only 2239 patients (60%) visited one of the FQHC clinics before and after referral. The availability of clinic visits and health outcomes, relative to the date of referral, is the first major limitation of our study, discussed in greater detail in Appendix Section A.8. Figure 2 presents a geographic overview of the study area. The randomized home addresses (± 500 m) of treated (referred) and control (not referred) patients are represented by black and gray colored points, respectively. Compliance is not color indicated. Cartographic features of the study area, such as major highways, are masked to preserve anonymity.

Patients' home addresses, food pantry locations, and primary care clinics demonstrate spatial clustering patterns across both rural and urban regions of the study area. The driving distance (in miles) between the patients' randomized home addresses and the nearest food pantry does not vary significantly. Therefore, proximity to the pantry does not necessarily explain compliance with the referral. Table 1 summarizes patient, FQHC clinic, and food pantry characteristics by referral and compliance (1+ redemptions). We required all patients to have sufficient data or three clinic health outcomes before referral, three after referral and within a 24-month window. The cut-off point of three has been used in other studies (Ahuja et al., 2020). For control patients, clinic visits before the program launch date of August 10, 2016 were classified as pre-intervention.

The final treated and control patient sample had at least six health outcomes with similar timing of clinic visits, relative to the same patients' date of referral (see Appendix Section A.1.1). *Compliers* had 1 or more food pantry visits in the 12-months after referral. *Noncompliers* had zero. *HighComorbid* compliers had a Charlson Comorbidity Index (CCI) of 3 or greater at the time of referral, and *HighPantryUse* compliers had 11 or more food pantry visits after referral. Consistent with other studies like Ranjit et al. (2023), we used one Pharmacy redemption as the threshold for compliance.⁷

The final sample of 1052 patients was composed of 306 compliers, 434 noncompliers and 312 control

TABLE 1 Summary statistics.

			# of food pantry visits 0–12-months after referral	
			0 redemptions ^a	1+ redemptions ^a
		Referred		
Patient population	22,930	3709	2327	1382
Patient characteristics				
Male, %	37.6	30.8	30.4	31.6
Race: non-Hispanic white, %	22.5	22.7	23.3	21.5
Race: non-Hispanic black, %	42.0	47.5	45.6	50.6
Age, < 29 years, %	13.9	6.9	6.7	7.4
Age, ≥ 60 years, %	23.4	27.8	27.9	27.6
Geolocation of randomized home address (±500 m), %	99.6	99.8	99.6	95.6
TransitScore™, mean ± SD	30.6 ± 13.2	31.8 ± 12.8	31.5 ± 12.8	32.1 ± 12.8
ADI Quintile, mean ± SD	3.9 ± 1.2	4.0 ± 1.2	4.0 ± 1.2	4.1 ± 1.2
FQHC clinic characteristics				
Clinic visits, n	97,774	22,400	13,733	8667
Patient-quarters, n	75,513	16,119	9935	6184
BMI, kg/m ² , mean ± SD	33.8 ± 8.66	32.9 ± 8.49	33.9 ± 8.92	34.5 ± 8.92
Missing, %	3.9	1.4	0.7	1.6
HgbA1c, %, mean ± SD	7.73 ± 1.89	7.77 ± 1.93	7.70 ± 2.22	7.67 ± 2.24
Missing, %	52.4	56.8	50.4	45.6
Charlson ≥2, %	3.1	5.6	5.2	6.1
Food pantry visits				
Number of pantry visits, n	10,536	10,536	0	10,536
Number of unique pantries, mean ± SD	2.32 ± 1.95	2.32 ± 1.95	0.232 ± 1.95	
One-way distance to nearest pantry, miles, mean ± SD	134.3 ± 254.3	7.8 ± 36.3	7.5 ± 15.1	8.5 ± 56.1

^aFarmacy “redemptions” is adapted from Ranjit et al. (2023)

patients. Control patients were selected to ensure comparable availability of clinical health outcome data to that of the complier sample, matched using sample statistics for *Days_elapsed* or the number of days between a clinic visit and referral (see Appendix Section A.1.1).

4.3 | Identification strategy

We use DiD to identify the effect of the Farmacy program on preventative health quality. To test the supporting assumptions of DiD, we aggregate data to the patient-quarter level of analysis and impute missing data to overcome some of the inherent limitations in the data (see Appendix Sections A.2 and A.6).

We use ordinary least squares and a hierarchical specification, which is built from a base model, indicated in Equation (1), with the inclusion of additional terms to constitute a heterogeneous effects model, indicated in Equation (2):

$$Y_{it} = \alpha_i + \beta_1 Post_{it} + \beta_2 Post_{it} \times NonComplier_i + \beta_3 Post_{it} \times Complier_i + \lambda_{it} + \epsilon_{it} \quad (1)$$

$$Y_{it} = \alpha_i + \beta_1 Post_{it} + \beta_2 Post_{it} \times NonComplier_i + \beta_3 Post_{it} \times Complier_i + \beta_4 Post_{it} \times Complier_i \times HighComorbid_i + \beta_5 Post_{it} \times Complier_i \times HighPantryUse_i + \beta_6 Post_{it} \times Complier_i \times HighComorbid_i \times HighPantryUse_i + \lambda_{it} + \epsilon_{it} \quad (2)$$

where Y_{it} is the preventative health quality outcome for patient i at the normalized clinic visit at time t , referral occurs at $t=0$, α_i is a patient fixed effect, $Post_{it}$ is a dummy variable that takes a value of 1 for all patient-clinic visits after referral, and 0 otherwise, λ_{it} is the number of days elapsed between the clinic visit at time t and referral, and ϵ_{it} is the error term.

Complier takes the value of 1 for patients that received a Farmacy referral and had at least one food

pantry visit 1 year after referral. The value is 0 for non-compliers and control patients. Control patients are those that were not referred to the Farmacy program. *NonComplier* takes a value of 1 for referred patients with zero food pantry visits, and a 0 for compliers and control patients. Compliers are further categorized by their health status (at the time of referral) and the level of food pantry use (after referral). Note that these variables, *HighComorbid* and *HighPantryUse*, are defined in a time-invariant and binary manner. *HighComorbid* takes a value of 1 for a complier with a CCI ≥ 3 , and 0 otherwise. CCI scores are based on all diagnosis codes across all clinic visits before and at the time of referral. *HighPantryUse* takes a value of 1 for a patient with ≥ 11 food pantry visits in the 1-year period after referral, and 0 otherwise. The coefficients β_4 , β_5 , and β_6 can be interpreted as the incremental effect of Farmacy participation for compliers with a CCI ≥ 3 , high pantry use after referral, and both, respectively. To reduce concern for time-invariant factors like demographics, patient fixed effects (α_i) are included in all model specifications. All time-invariant variables including *Noncomplier_i*, *Complier_i*, *HighComorbid_i* and *HighPantryUse_i* and their corresponding time-invariant

interactions are absorbed by α_i and “difference out” of the model.

5 | RESULTS

We begin by examining the evidence for H1. In Table 2, we find that the coefficient for *Post* $\times$ *Noncomplier* is positive and significant across models (2)–(4). This suggests that patients who screened positive for food insecurity and were referred to the Farmacy but did not visit the food pantry had a significant increase in BMI after referral, which is consistent with the facts that obesity is a progressive disease and food insecurity is a risk factor for health. The coefficient for *Post* $\times$ *Complier* ranges from 0.12 to 0.62, which is smaller in magnitude and not statistically significant. This BMI stabilization effect is observed only in the complier sample, which suggests that the Farmacy program helped patients to better manage their health.

We also find evidence of a potential dose–response relationship between pantry use and health outcomes. First, we adjust the complier sample to include those

TABLE 2 The effect of the Farmacy program on BMI and HbA1c.

	BMI				HbA1c
	(1)	(2)	(3)	(4)	(5)
<i>Post</i>	−0.08** (0.03)	−0.31** (0.13)	−0.36** (0.14)	−0.31*** (0.10)	−0.09 (0.09)
<i>Post</i> $\times$ <i>NonComplier</i>	0.07 (0.07)	0.30** (0.13)	0.67* (0.36)	0.26*** (0.09)	0.06 (0.08)
<i>Post</i> $\times$ <i>Complier</i>	0.21 (0.22)	0.50 (0.33)	0.62 (0.39)	0.12 (0.39)	0.26 (0.35)
<i>Post</i> $\times$ <i>Complier</i> $\times$ <i>HighComorbid</i>				0.90*** (0.19)	−0.18 (0.20)
<i>Post</i> $\times$ <i>Complier</i> $\times$ <i>HighPantryUse</i>				0.43 (0.34)	−0.34 (0.30)
<i>Post</i> $\times$ <i>Complier</i> $\times$ <i>HighComorbid</i> $\times$ <i>HighPantryUse</i>				−1.49** (0.60)	−1.01* (0.52)
Patient FEs	✓	✓	✓	✓	✓
Observations	93,962	9029	3206	9029	3397
Patients	22,930	1052	397	1052	718
Adjusted R^2	0.02	0.45	0.62	0.65	0.74

Note: Huber-white standard errors in parentheses. Model (2) limits the control sample to $n = 312$ patients using Days elapsed, as outlined in Appendix Section A.1.1. This ensures that treated patients are compared to similar control patients based on demographic characteristics and the availability of health outcome data. Model (3) adjusts the complier sample and requires a complier to have 6 or more pantry visits. A similar matching procedure is used to adjust the noncomplier sample and ensure similar availability of health outcome data across compliers and noncompliers (e.g., *Days elapsed* SD ≥ 80 , minimum ≤ 60 and maximum ≥ 200). This sample includes $n = 39$ compliers, $n = 46$ noncompliers and $n = 312$ control patients. Model (4) and (5) use Equation (2) and the cohort from model (2).

* $p < .10$; ** $p < .05$; *** $p < .01$.

with at least 6 pantry visits after referral. Then we use *Days_elapsed* (see Appendix Section A.1.1) to characterize the timing of clinic visits in this patient cohort. We use multiple summary statistics for *Days_elapsed*, including standard deviation (SD), min and max, to select a representative comparison *Noncomplier* sample, combined with the original control sample. The results are provided in model (3). We find that the coefficient for $Post \times Noncomplier$ increases in magnitude and is marginally significant, while the coefficient for $Post \times Complier$ remains insignificant. Both model (2) and model (3) show that BMI increased for noncompliers after referral. However, this increase appears to be larger in magnitude for noncompliers when compared to compliers with at least 6 pantry visits and the same control patients.

Second, using the same complier sample from model (2), we introduce the $HighComorbid_i$ and $HighPantryUse_i$ variables, according to the heterogeneous effects model in Equation (2). β_6 is negative and significant, with an absolute magnitude larger than β_3 in model (2), suggesting that high comorbid and high pantry use compliers had a greater improvement in health. In other words, high comorbid compliers with ≥ 11 food pantry visits after referral appeared to have a greater reduction in BMI and HbA1c, relative to those with 1 or more food pantry visits. This cohort also experienced a marginally significant reduction in HbA1c. The total effect—or the sum of the two marginal effects of $Post \times Complier$ and

$Post \times Complier \times HighComorbid \times HighPantryUse$ —of the Farmacy program on BMI (HbA1c) is -1.37 (-0.75).

To make sense of the clinical and cost significance of these findings, the average patient in our complier cohort had a HbA1c close to 6.9 at the time of referral and would therefore be classified as “diabetes-range A1c” (e.g., $HbA1c \geq 6.5$). It is generally accepted that a 0.5% reduction in HbA1c is clinically meaningful, and a full 1% reduction in HbA1C has been associated with a 37% lower odds of developing microvascular complications (Stratton et al., 2000). Importantly, a 1% reduction in HbA1C has been associated with a 2% reduction in all-cause total healthcare costs and a 13% reduction in diabetes-related total healthcare costs (\$429 and \$736 per year in 2017 dollars, respectively) (Lage & Boye, 2020). The partnership between healthcare and foodbanks has the potential to substantially impact the health outcome and healthcare costs of the most vulnerable populations, purely through ensuring regular access to food. In Appendix Section A.3, we find that the magnitude of the local average treatment effect on HbA1c for high comorbid and high pantry use compliers is close to 2%. The combined evidence supports H1.

5.1 | Robustness checks

In this section, we summarize results of the robustness checks used to validate the main effect estimate and undertake an extensive qualitative exploration of the Farmacy program to reduce concern for other possible behaviors that could bias our results.

5.1.1 | Falsification test

We perform a falsification by using “placebo” in a contrived sample that employs randomized group assignment. The placebo test falsely assigns high pantry use and high comorbidity to a similar proportion of control patients and re-estimates the DiD coefficients. Coefficients as extreme or more extreme than the ones observed in our main analysis would suggest that another mechanism may be driving the observed effect. More information is provided in Appendix Section A.5. The test supports our main results.

5.1.2 | Features of high pantry use

First, we find that patient demographics do not show a significant correlation with pantry use (Appendix Section A.7). Next, among high pantry use compliers—a cohort of patients for which sufficient pantry data is available—we observed no significant differences in the nutritional content of the food received (Appendix Section A.8). We did find a difference in longitudinal access patterns. Appendix Section A.9 explores differences in food pantry search, frequency, and consistency among high pantry use compliers that are associated with low and high comorbid status. High comorbid, high pantry use patients appeared to rely on the Farmacy program as intended or use “food as medicine.”

5.1.3 | Qualitative exploration of the Farmacy program

We corroborate our analyses with a qualitative discussion of the Farmacy program. We held a number of focus group discussions with food bank practitioners and food pantry volunteers to better understand food pantry access behaviors in the Farmacy program. Focus groups were scheduled on a semi-monthly basis starting in September 2018. From October to December 2018, we conducted semi-structured interviews with three healthcare providers from three FQHC clinics and the lead dietitian at the MOFC. All interviews were arranged and coordinated

by a manager at MOFC and conducted in-person. We present our findings as a series of questions and answers in the Appendix Section A.10.

6 | CLINIC UTILIZATION

In this section, we examine the evidence for H2. We also transition from the patient-visit level of analysis to a clinic-day level of analysis to explore changes in clinic utilization across the three cohorts. We capture the total number of clinic encounters each day for referred non-compliers and control nonparticipants. This enables us to examine differences in clinic utilization by food security status given that noncompliers were referred to the program and food insecure, while nonparticipants were not referred and food secure (see Appendix Section A.4).

In the first part of the analysis, the sample is limited to noncompliers and nonparticipants, and clinic volume categorized accordingly to capture the total volume of food insecure and food secure patients for a given clinic day. Clinic volume is explained by day-of-week, week, month, year, and clinic dummy variables. To examine temporal variation in clinic visit volume by food security status, we interact day-of-week with noncomplier using the following specification,

$$\begin{aligned} \text{LOG}(\text{ClinicVol}_{jt}) = & \beta_0 + \sum_{j=1}^5 \beta_j \text{Day}_{jt} \\ & + \underbrace{\sum_{j=1}^5 \beta_j \text{Day}_{jt} * \text{Noncomplier}_{jt}}_{\text{Model (3) only}} \\ & + \sum_{j=1}^{52} \alpha_j \text{Week}_{jt} + \sum_{j=1}^{12} \gamma_j \text{Month}_{jt} \\ & + \sum_{j=2015}^{2018} \theta_j \text{Year}_{jt} + \sum_{j=1}^8 \tau_j \text{Clinic}_j \\ & + \epsilon_{jt} \end{aligned} \quad (3)$$

where $\text{LOG}(\text{ClinicVol}_{jt})$ is the log-volume of total clinic encounters for clinic j on day t and Day_{jt} is a set of dummy variables representing the day of the week. Monday is the first day of the week and tends to have high clinic volume (see Figure A4). Thus, we use Monday as the reference category. All clinics are closed on Sunday. None of the clinics changed its operating hours throughout the study period. Week_{jt} , Month_{jt} , and Year_{jt} are sets of dummy variables representing the numeric week, month, and year of the clinic visit, respectively. We include these controls to adjust for expected seasonality in clinic visits. Clinic_i is a set of dummy variables for the

eight FQHC clinic locations. We remove the month of December since utilization patterns likely change during the winter holidays (Byrne & Just, 2021).

To characterize differences in pantry utilization by day of the week, we first classify clinic volume using the binary variable Noncomplier_{jt} . This variable is slightly different than the Noncomplier_i variable used in the main DiD, where it is used to classify patients. Here we use it to classify clinic volume. Noncomplier_{jt} is set to 1 for clinic visits by food insecure noncompliers and 0 for clinic visits by food secure nonparticipants (i.e., clinic volume from those not referred or food secure). Complier visits are excluded in the first round of analysis. Later, we examine variation in compliers' clinic volume before referral and after referral. We use these two distinct samples of complier clinic visits and Equation (3) without the interaction term. We compare the significance of the day-of-week coefficients for compliers before and after referral to make inferences on the possible effect of the Farmacy program on health behaviors.

Table 3 reveals several interesting patterns related to clinic visitations. First, the probability of food secure nonparticipants and food insecure noncompliers visiting the clinic diminishes as the week progresses. This is also observed in Figure A4. In models (1) and (2), the coefficients for day-of-week become more negative as the week progresses toward Saturday, suggesting that both groups have a lower probability of visiting the clinic later in the week. However, model (3) shows that food insecure non-compliers have a significantly higher probability of visiting the clinic on Fridays and Saturdays, relative to food secure nonparticipants. This may be due to competing demands, since food insecure adults must contend with other social factors that are known to inhibit health behaviors like, for example, their ability to attend a primary care appointment. For example, food insecure adults may be more likely to make clinic visits later in the week, as these days have fewer occupational demands.

With clinic utilization patterns by food security status firmly established from models (1)–(3), we turn our attention to clinic utilization for compliers before and after referral. In Table 3, model (4) shows little to no intra-week variation in clinic utilization among compliers before referral; notably, all day-of-week coefficients are non-significant. Therefore, before referral, compliers have no difference in the number of clinic visits by day of the week. However, after referral, clinic utilization changes. Model (5) shows a negative and significant coefficient for Saturday ($\beta = -0.50$, $p < .05$), reflecting a lower probability of a clinic visit. A lower probability of a clinic visit later in the week is consistent with the clinic utilization patterns observed among patients that were food secure: the coefficient for $\text{Saturday} \times \text{Noncomplier}$ in model (3) is positive and significant, meaning that food

TABLE 3 Day-of-week coefficients for clinic volume for food secure nonparticipants (model 1), food insecure noncompliers (2) and compliers before (4) and after referral (5).

	LOG(ClinVol)				
	(1)	(2)	(3)	(4)	(5)
<i>Tuesday</i>	−0.01 (0.04)	−0.03 (0.03)	−0.01 (0.03)	−0.00 (0.02)	−0.01 (0.03)
<i>Wednesday</i>	−0.12*** (0.04)	−0.08*** (0.03)	−0.12*** (0.03)	−0.02 (0.02)	−0.01 (0.03)
<i>Thursday</i>	−0.16*** (0.04)	−0.08*** (0.03)	−0.16*** (0.03)	−0.02 (0.03)	0.01 (0.04)
<i>Friday</i>	−0.21*** (0.04)	−0.12*** (0.03)	−0.21*** (0.03)	−0.01 (0.03)	−0.05 (0.04)
<i>Saturday</i>	−1.58*** (0.17)	−0.83*** (0.21)	−1.61*** (0.15)	−0.08 (0.22)	−0.50** (0.20)
<i>Tuesday</i> ×Noncomplier			−0.02 (0.05)		
<i>Wednesday</i> ×Noncomplier			0.04 (0.05)		
<i>Thursday</i> ×Noncomplier			0.07 (0.05)		
<i>Friday</i> ×Noncomplier			0.09* (0.05)		
<i>Saturday</i> ×Noncomplier			0.82*** (0.29)		
Week, Month, Year Controls	✓	✓	✓	✓	✓
Clinic FEs	✓	✓	✓	✓	✓
R ²	0.04	0.07	0.30	0.08	0.09
Observations	4584	3740	8288	2327	1317

Note: Robust standard errors clustered at the clinic level are in parentheses. Models (1) includes all clinic visits from food secure nonparticipants; Models (2) food insecure noncompliers; Model (3) includes both. Models (4) and (5) use the complier sample only. Model (4) includes all clinic visits before referral, while Model (5) includes all clinic visits after referral.

* $p < .1$; ** $p < .05$; *** $p < .01$.

insecure patients had a greater probability of visiting the clinic on Saturday, relative to food secure patients. This evidence suggests that compliers had clinic visit patterns similar to that of the food secure group, but only after referral. Pharmacy compliance may have helped patients better prioritize their care and attend primary care appointments early in the week. The change in clinic utilization before and after referral supports H2.

7 | DISCUSSION AND CONCLUSIONS

Partnership model of care represent a new supply chain approach to health promotion and disease

prevention and these models should be particularly well-suited to address the needs of patients with resource-constraints, who struggle to manage complex, long-term, chronic conditions (Jónasson et al., 2022). Our results suggest that partnership models may help to curb the risk of metabolic conditions and facilitate behavioral changes that enable greater stability in health outcomes. In the United States, chronic diseases represent 90% of the \$4.1 trillion in annual healthcare spend (CDC, 2023) and the prevalence of obesity is only expected to worsen over the next decade (Ward et al., 2019). Food and health partnership models offer tremendous potential. In this section, we outline our contributions and propose several areas of future research.

7.1 | Contributions

Our research makes four contributions to the HOM literature. First, we respond to the call for research beyond the hospitals' boundaries to include external collaboration (Adepoju et al., 2023; Ramdas et al., 2012). Although collaboration has been studied in healthcare, most of the research examines hospital-focused collaboration across providers (Lan et al., 2022). Extending the healthcare ecosystem to include new entities like CBOs creates opportunities for collaborative healthcare delivery. This study also represents a response to a new style of healthcare research, which aligns closely to what is referred to as "HOM 2.0" in Dai and Tayur (2020). Healthcare systems should continue to search for new ways to collaborate with others (KC et al., 2020; Keskinocak & Savva, 2020) and address patients' resource-limitations (Jónasson et al., 2022). The food and health partnership model is just a start.

Our study finds that high comorbid patients had greater search, frequency, and consistency of food pantry use, which suggests that when patients use "food as medicine", they have better health outcomes. This implies that health systems should partner with CBOs to address their patients' SDOH, especially for patients that require more complex care. This research extends previous work on access-based interventions focused on the availability and accessibility of resources. This literature reveals that access improves outcomes when individuals understand and ascribe value to the resource (Allcott et al., 2019; Calmon et al., 2018; Levi et al., 2019). External collaboration and integration with non-traditional health entities could potentially address access and education, simultaneously. How to link or route patients between the healthcare clinic and food pantry remains a central operational challenge and more research is needed in this area.

Second, our research strengthens the stance that patients play an important role in preventative and population-based models of care. According to Roels and Staats (2021), the patient cannot be considered an exogenous entity in healthcare delivery, but instead shares the accountability for their care (Johnston, 2019). Healthcare systems should redesign their care delivery processes to encourage and assist patients in being self-accountable to their health outcomes. External collaboration enables healthcare systems to address the patients' SDOH and facilitates their role as a co-producer of care (Bendoly & Oliva, 2024).

Healthcare providers can also engage the patient in a different manner. Repeat exposure and encouragement can increase compliance and strengthen behavioral commitments (Staats et al., 2017). People-centric factors

during the patient-provider relationship, to include things like gender concordance (Greenwood et al., 2018), could also play an important role in referral compliance, long-term behavior change and overall engagement with several components of care. In partnership models, the patient-provider relationship works in tandem with the patient-volunteer relationship (at the food pantry) to affect the patients' health behaviors. These relationships deserve more attention, especially when the patient is located in a rural, hard-to-reach area.

Third, geographic proximity had limited effect on Pharmacy compliance. Instead, we find that the patient's health status at the time of referral may be a better indicator of the number of Pharmacy redemptions. Among high pantry use compliers, high comorbid patients had more consistent use of the food pantry and, based on a geographic analysis of compliance, these patients also appeared to be willing to travel further from their home address to visit the food pantry. Since greater travel costs correspond to a greater valuation of the service (Byrne & Just, 2023), high comorbid patients may have valued the food pantry visit more than their low comorbid counterparts. It could also be the case, however, that these patients devised more creative strategies to visit the food pantry, such as piggybacking food pantry visits off of other planned trips. Geographic factors and unobserved pantry visit trip information make it difficult to study compliance in partnership models of care, especially since individuals vary in their travel and commuting patterns and these differences have important implications for their food environment (García Bulle Bueno et al., 2024). To better conceptualize the integration between healthcare providers and community food systems, new models of food access rely on mobile distribution of food (Stauffer et al., 2022), which circumvent common access barriers associated with fixed-location food pantries.

Finally, our qualitative interviews suggest that food and health partnership models should expand the scope of resources offered and add concurrent features to facilitate compliance (DePuccio et al., 2022). For example, a recent study by Palakshappa et al. (2023) finds that food insecurity affects the entire family. Therefore, partnership models may change to reflect more of a family-based therapy approach to include other programmatic elements like cooking classes that educate patients—and their families—on how to properly prepare, store, and consume healthy foods.

In review, health systems should use partnership models to ensure patients continue to remain healthy outside of their care boundaries. These types of models are gaining more traction, as recently evidenced by a partnership in the greater Cleveland area between

Cleveland Clinic, University Hospitals, MetroHealth and Greater Cleveland Food Bank (Kelm, 2023). Considering the totality of interactions between health systems, patients and the community holds considerable promise and helps illuminate how operational changes impact outcomes (Bendoly & Oliva, 2024).

7.2 | Limitations and future research

We acknowledge a number of limitations of our study and suggest new avenues for future research. First, our study relies on retrospective, observational data, meaning that it is possible that our results were confounded. Future studies should consider randomization to improve causal inference and better isolate the clinical benefits of eating healthy (Doyle et al., 2023; Moran & Roberto, 2023). Such studies may help resolve which partner is ultimately responsible for financing social risk interventions (Berkowitz et al., 2023). Overall, more research is needed to understand how SDOH like food insecurity can be addressed in healthcare delivery in a systematic and equitable manner (Sunar & Swaminathan, 2022, p. 4384).

Second, volunteer-patient interactions at the food pantry seemed important to the success of the program and peer learning likely occurred across consecutive visits. However, we did not collect granular patient, provider, and volunteer interaction data. Collecting interaction-specific data through patient surveys or monitoring technologies might reveal important people-centric contingencies of these programs. There are also patient-to-patient interactions at the food pantry, which could further enhance health behaviors in a manner similar to shared medical appointments (Ramdas & Darzi, 2017). Peer modeling and peer learning encourage behavior change and might displace or eliminate clinical interactions altogether. Understanding the peer-learning effect could inform efforts to scale up these programs.

Notwithstanding these limitations, our study demonstrates that partnership models of care can improve health outcomes and behaviors. As healthcare systems continue to change and evolve, partnership-based approaches can help to shape and sustain healthy behaviors at the population level, especially if the partnership includes multiple CBOs and offers a greater breadth of health interventions (CMS.gov, 2023). The Farmacy program that we studied showed that patients make use of the referral in interesting and non-trivial ways, but another support resource such as a care coordinator could increase compliance by overseeing and managing referrals. Partnership models represent a scalable solution to preventative health issues and promote behavior

change among low-income patients with chronic health conditions.

Future research should focus on the role of volunteer-patient interactions in partnership models of care, which could improve coordination and concordance. CBO volunteers are more likely than healthcare providers to share patients' neighborhoods and SES, which can translate to better communication and information sharing, and shared experiences and group identities can result in better health outcomes (Greenwood et al., 2018). Concordance with CBO volunteers could also help to mitigate implicit bias in healthcare (Salles et al., 2019) and engender the higher levels of trust that are critical in the context of long-term care. Another way to create trust is through peer mentor relationships, which have been successfully implemented at hospital discharge (Chun et al., 2022). We propose that pantry volunteers, who may have struggled in the past with food insecurity, could serve others in a role similar to a peer mentor and support patients' access to and use of preventative health services.

In the aggregate, our research shows that partnership models offer a low-cost, scalable opportunity for healthcare systems to improve the health of low-income patients struggling with chronic metabolic conditions. Food and health partnership models are also a low-risk intervention, as food pantries are nearly ubiquitous and most low-income patients would benefit from more reliable and consistent access to healthy food. The Farmacy program also represents a timely approach to counter to rising rates of obesity (Ward et al., 2019) and community referrals could help address multiple causes of chronic diseases. For example, if a CBO visit acts as a substitute for a clinic visit, or at least reduces inappropriate use of hospital resources, there may also be important, yet understudied effects on clinical productivity. Patients that redeem their referral may have a longer clinic revisit interval or shorter visits, which decrease provider utilization. Better health outcomes with lower provider utilization could revolutionize healthcare delivery.

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ENDNOTES

- ¹ SDOH are the conditions in which people are born, grow, live, work and age (Hill-Briggs et al., 2021) and include the non-clinical aspects of health, such as food security, financial literacy and access to reliable transportation. Social factors are responsible for up to 80% of the patients' health outcomes (Hussein & Collins, 2016).

- ² The challenge of cross-sector data sharing and, in many cases, the lack of resources or capabilities on the part of the non-profit CBO are two other limiting factors (Walker et al., 2022).
- ³ The MOFC is the largest Feeding America member food bank in the state of Ohio with a service territory that stretches across 20 counties. In 2022, the MOFC distributed close to 80 million pounds of food or more than 100,000 meals daily (MOFC Annual Report 2022).
- ⁴ Individuals in Ohio with an income $\leq 135\%$ FPL are eligible for public food assistance programs like the Supplemental Nutrition Assistance Program (SNAP). Although we do not have this data, 80% of the FQHC patient population, being at or below 100% of the FPL, were likely enrolled in SNAP. Low-income patients at the FQHC with SNAP benefits during the study period would likely have screened negative in response to the clinic-based food security screening.
- ⁵ Partnerships can also include multiple entities. In the expanded chronic care model, *multiple CBOs* triage health-related social needs (Barr et al., 2003). One example, the Community HUB model, employs distinct social service pathways (<https://pchihub.com/>) to address a number of SDOH. The Pharmacy program is a simplified version of a chronic care model in which clinics only partner with the food bank.
- ⁶ Feeding America estimates the incidence of food insecurity to be 12.7%. (<https://map.feedingamerica.org/county/2019/overall/ohio/organization/mid-ohio-foodbank>).
- ⁷ Pantry use is extremely low among low-income patients in clinic-based food referral programs. Ranjit et al. (2023) finds that “the distribution of redemptions is positively skewed” (p. 4). 85% of referred patients made 12 or fewer pantry visits, and 10% more than 18 visits. For our study, the distribution of the number of pantry visits is shown in Figure A2.
- ⁸ Although using the data reduction steps to address the first-order sample size limitations results in a more balanced cohort of treated and control patients, each patient in the cohort still has a considerable amount of missing data. The corresponding data steps require an included patient have three health outcomes before and after referral or six data points over a two-year period.
- ⁹ The total effect estimate for a food insecure patient visiting the clinic on a Saturday is $= -1.61 + 0.82 = -0.8$, and the odds are, $e^{-0.8} - 1 = 66\%$.
- ¹⁰ Clients who visit food pantries receive a standard allocation of food adjusted for household size. Enough food is provided to support a client and his or her family for a two-week period.
- ¹¹ Since noncompliers and control patients made zero food pantry visits, we cannot examine differences in food pantry use by food security status.
- ¹² Round trip distance was calculated for each visit to the food pantry. A visit was assumed to have originated from the patient's home address. Averages were first calculated within patient (e.g., one patient can visit many pantries) and then across patients within a particular group (e.g., low and high comorbid compliers).
- ¹³ It is interesting to note that during the time frame of our study, our partnering food bank changed its name to “Mid-Ohio Food Collective” to minimize the stigma associated with food bank.

REFERENCES

- Adepoju, T. A., Tucker, A., Long, P., & Manasseh, C. (2023). *External collaborative coordination: Improving efficiency in a dedicated observation unit*. Working Paper.
- Adler, N. E., & Newman, K. (2002). Socioeconomic disparities in health: Pathways and policies. *Health Affairs*, 21(2), 60–76.
- Ahuja, V., Alvarez, C. A., & Staats, B. R. (2020). Maintaining continuity in service: An empirical examination of primary care physicians. *Manufacturing & Service Operations Management*, 22(5), 1088–1106.
- Ahuja, V., Alvarez, C. A., & Staats, B. R. (2022). An operations approach for reducing glycemic variability: Evidence from a primary care setting. *Manufacturing & Service Operations Management*, 24(3), 1474–1493.
- Allcott, H., Diamond, R., Dub'e, J. P., Handbury, J., Rahkovsky, I., & Schnell, M. (2019). Food deserts and the causes of nutritional inequality. *The Quarterly Journal of Economics*, 134(4), 1793–1844.
- Anderson, D., Gao, G., & Golden, B. (2014). Life is all about timing: An examination of differences in treatment quality for trauma patients based on hospital arrival time. *Production and Operations Management*, 23(12), 2178–2190.
- Andrews, I., Stock, J. H., & Sun, L. (2019). Weak instruments in instrumental variables regression: Theory and practice. *Annual Review of Economics*, 11, 727–753.
- Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American statistical Association*, 91(434), 444–455.
- Barr, V., Robinson, S., Marin-Link, B., Underhill, L., Dotts, A., Ravensdale, D., & Salivaras, S. (2003). The expanded chronic care model. *Hospital Quarterly*, 7(1), 73–82.
- Batt, R. J., & Terwiesch, C. (2015). Waiting patiently: An empirical study of queue abandonment in an emergency department. *Management Science*, 61(1), 39–59.
- Baum, A., Garofalo, J. P., & Yali, A. M. (1999). Socioeconomic status and chronic stress: Does stress account for SES effects on health? *Annals of the New York Academy of Sciences*, 896(1), 131–144.
- Bavafa, H., Canamucio, A., Marcus, S. C., Terwiesch, C., & Werner, R. M. (2022). Capacity rationing in primary care: Provider availability shocks and channel diversion. *Management Science*, 68(4), 2842–2859.
- Bavafa, H., Hitt, L. M., & Terwiesch, C. (2018). The impact of e-visits on visit frequencies and patient health: Evidence from primary care. *Management Science*, 64(12), 5461–5480.
- Bavafa, H., Savin, S., & Terwiesch, C. (2019). Managing patient panels with nonphysician providers. *Production and Operations Management*, 28(6), 1577–1593.
- Bendoly, E., & Oliva, R. (2024). Holding North: Recognizing identity and advancing contribution in operations management. *Journal of Operations Management*, 518–522.
- Berkowitz, S. A., Basu, S., Meigs, J. B., & Seligman, H. K. (2018). Food insecurity and health care expenditures in the United States, 2011–2013. *Health Services Research*, 53(3), 1600.
- Berkowitz, S. A., Gottlieb, L. M., & Basu, S. (2023). Financing health care system interventions addressing social risks. In *JAMA health forum* (Vol. 4, No. 2, pp. e225241–e225241). American Medical Association.

- Berkowitz, S. A., Parashuram, S., Rowan, K., Andon, L., et al. (2018). Association of a care coordination model with health care costs and utilization: The Johns Hopkins Community Health Partnership (J-CHiP). *JAMA Network Open*, 1(7), e184273.
- Berkowitz, S. A., Seligman, H. K., & Choudhry, N. K. (2014). Treat or eat: Food insecurity, cost-related medication underuse, and unmet needs. *American Journal of Medicine*, 127(4), 303–310.
- Berkowitz, S. A., Seligman, H. K., Meigs, J. B., & Basu, S. (2018). Food insecurity, healthcare utilization, and high cost: A longitudinal cohort study. *The American Journal of Managed Care*, 24(9), 399.
- Berkowitz, S. A., Terranova, J., Hill, C., Ajayi, T., et al. (2018). Meal delivery programs reduce the use of costly health care in dually eligible Medicare and Medicaid beneficiaries. *Health Affairs*, 37(4), 535–542.
- Betcheva, L., Erhun, F., & Jiang, H. (2020). OM Forum—Supply chain thinking in healthcare: Lessons and outlooks. *Manufacturing & Service Operations Management*, 23(6), 1333–1353.
- Bolandifar, E., DeHoratius, N., Lennon Olsen, T., & Wiler, J. L. (2016). Modeling the behavior of patients who leave the emergency department without being seen. *Chicago Booth Research Paper*, 14-12, 430–446.
- Bonney, J., & Chang, D. (2018). Community Care Coordination Systems: Connecting Patients to Community Services. <https://www.movinghealthcareupstream.org/wp-content/uploads/2018/06/community-care-coordination-systems.pdf>
- Brody, J. (2018). *The best reason to watch your waistline*. Herald Tribune. <https://www.heraldtribune.com/entertainmentlife/2018/0619/brody-best-reason-to-watch-your-waistline>
- Byrne, A. T., & Just, D. R. (2021). The other half: An examination of monthly food pantry cycles in the context of SNAP benefits. *Applied Economic Perspectives and Policy*, 43(2), 716–731.
- Byrne, A. T., & Just, D. R. (2023). What is free food worth? A non-market valuation approach to estimating the welfare effects of food pantry services. *American Journal of Agricultural Economics*, 105(4), 1063–1087.
- Calmon, A. P., Rajasingh, D. J., Romero, G., & Stenson, J. (2018). *Operations strategy at the base of the pyramid: Consumer education and reverse logistics in a durable goods supply chain*. INSEAD.
- CDC. (2023). Health and Economic Cost of Chronic Disease. <https://www.cdc.gov/chronicdisease/about/costs/index.htm>
- Certo, S. T., Busenbark, J. R., Woo, H. S., & Semadeni, M. (2016). Sample selection bias and Heckman models in strategic management research. *Strategic Management Journal*, 37(13), 2639–2657.
- Choi, B. Y. (2021). Instrumental variable estimation of truncated local average treatment effects. *PLoS One*, 16(4), e0249642.
- Chun, Y., Harris, S. L., Chandrasekaran, A., & Hill, K. (2022). Improving care transitions with standardized peer mentoring: Evidence from intervention based research using randomized control trial. *Journal of Operations Management*, 68(2), 185–214.
- Clark, A., Walker, D., & Headings, A. (2020). *Addressing food insecurity in clinical care: Lessons from the mid-Ohio Pharmacy experience*. Health Affairs Forefront.
- CMS.gov. (2023). Accountable Health Communities. <https://innovation.cms.gov/initiatives/ahcm/>
- Dai, T., & Tayur, S. (2020). Om Forum—Healthcare operations management: A snapshot of emerging research. *Manufacturing & Service Operations Management*, 22(5), 869–887.
- De Marchis, E. H., Torres, J. M., Benesch, T., Fichtenberg, C., et al. (2019). Interventions addressing food insecurity in health care settings: A systematic review. *The Annals of Family Medicine*, 17(5), 436–447.
- Deloitte. (2019). 2019 Global health care outlook Shaping the future. <https://www2.deloitte.com/content/dam/Deloitte/global/Documents/Life-Sciences-Health-Care/gx-lshc-hc-outlook-2019.pdf>
- DePuccio, M. J., Garner, J. A., Hefner, J. L., Coovert, N., Clark, A., & Walker, D. M. (2022). Multi-stakeholder perspectives on the implementation of a clinic-based food referral program for patients with chronic conditions: A qualitative examination. *Translational Behavioral Medicine*, 12(9), 927–934.
- Dobrzykowski, D. D., & Tarafdar, M. (2015). Understanding information exchange in healthcare operations: Evidence from hospitals and patients. *Journal of Operations Management*, 36, 201–214.
- Doyle, J., Alsan, M., Skelley, N., Lu, Y., & Cawley, J. (2023). Effect of an intensive food-as-medicine program on health and health care use: A randomized clinical trial. *JAMA Internal Medicine*, 184(2), 154–163.
- DPP. (2002). The diabetes prevention program (DPP): Description of lifestyle intervention. *Diabetes Care*, 25(12), 2165–2171.
- Escobar, E. R., Pathak, S., & Blanchard, C. M. (2021). Peer reviewed: Screening and referral care delivery services and unmet health-related social needs: A systematic review. *Preventing Chronic Disease*, 18, 1–22.
- Feinberg, A. T., Hess, A., Passaretti, M., & Lee, T. H. (2018). Prescribing food as a specialty drug. *NEJM Catalyst*, 4(2).
- García Bulle Bueno, B., Horn, A. L., Bell, B. M., Bahrami, M., Bozkaya, B., Pentland, A., de la Haye, K., & Moro, E. (2024). Effect of mobile food environments on fast food visits. *Nature Communications*, 15(1), 2291.
- Garcia, S. P., Haddix, A., & Barnett, K. (2018). Peer reviewed: Incremental health care costs associated with food insecurity and chronic conditions among older adults. *Preventing Chronic Disease*, 15, 1–11.
- Greenwood, B. N., Carnahan, S., & Huang, L. (2018). Patient-physician gender concordance and increased mortality among female heart attack patients. *Proceedings of the National Academy of Sciences*, 115(34), 8569–8574.
- Gubler, T., Larkin, I., & Pierce, L. (2018). Doing well by making well: The impact of corporate wellness programs on employee productivity. *Management Science*, 64(11), 4967–4987.
- Hager, E. R., Quigg, A. M., Black, M. M., Coleman, S. M., et al. (2010). Development and validity of a 2-item screen to identify families at risk for food insecurity. *Pediatrics*, 126(1), e26–e32.
- Hassan, S. M., Giebel, C., Morasae, E. K., Rotheram, C., Mathieson, V., Ward, D., Reynolds, V., Price, A., Bristow, K., & Kullu, C. (2020). Social prescribing for people with mental health needs living in disadvantaged communities: The life rooms model. *BMC Health Services Research*, 20(1), 1–9.
- Helms, N. (2019). Study: CommunityRx helps patients gain confidence to find, navigate nearby health resources. <https://www.uchicagomedicine.org/forefront/community-articles/2019/february/communityrx-helps-patients-gain-confidence>

- Hibbard, J. H., & Greene, J. (2013). What the evidence shows about patient activation: Better health outcomes and care experiences; fewer data on costs. *Health Affairs*, 32(2), 207–214.
- Hill-Briggs, F., Adler, N. E., Berkowitz, S. A., Chin, M. H., et al. (2021). Social determinants of health and diabetes: A scientific review. *Diabetes Care*, 44(1), 258–279.
- Hopp, W. J., Li, J., & Wang, G. (2018). Big data and the precision medicine revolution. *Production and Operations Management*, 27(9), 1647–1664.
- Hussein, T., & Collins, M. (2016). The community cure for health care. *Stanford Social Innovation Review*. https://ssir.org/articles/entry/the_community_cure_for_health_care
- Hwang, J., & Christensen, C. M. (2008). Disruptive innovation in health care delivery: A frame-work for business-model innovation. *Health Affairs*, 27(5), 1329–1335.
- Johnston, S. C. (2019). The risk and cost of limited clinician and patient accountability in health care. *JAMA*, 322(18), 1759–1760.
- Jónasson, J. O., Ramdas, K., & Sungu, A. (2022). Social impact operations at the global base of the pyramid. *Production and Operations Management*, 31(12), 4364–4378.
- KC, D. S., Scholtes, S., & Terwiesch, C. (2020). Empirical research in healthcare operations: Past research, present understanding, and future opportunities. *Manufacturing & Service Operations Management*, 22(1), 73–83.
- KC, D. S., & Terwiesch, C. (2011). The effects of focus on performance: Evidence from California hospitals. *Management Science*, 57(11), 1897–1912.
- Kelm, A. (2023). Community Partners Come Together Pledging Further Commitment to Addressing Hunger in Greater Cleveland. <https://news.uhhospitals.org/News-Releases/articles/2023/12/community-partners-come-together-to-to-pledge-further-commitment-to-addressing-hunger>
- Keskinocak, P., & Savva, N. (2020). A review of the healthcare-management (modeling) literature published in manufacturing service operations management. *Manufacturing & Service Operations Management*, 22(1), 59–72.
- Kim, S. H., Chan, C. W., Olivares, M., & Escobar, G. (2015). ICU admission control: An empirical study of capacity allocation and its implication for patient outcomes. *Management Science*, 61(1), 19–38.
- Kohnke, E. J., Mukherjee, U. K., & Sinha, K. K. (2017). Delivering Long[U+2010] term surgical Care in Underserved Communities: The enabling role of international NPOs as partners. *Production and Operations Management*, 26(6), 1092–1119.
- Lage, M. J., & Boye, K. S. (2020). The relationship between HbA1c reduction and healthcare costs among patients with type 2 diabetes: Evidence from a U.S. claims database. *Current Medical Research and Opinion*, 36(9) 1441–1447.
- Lan, Y., Chandrasekaran, A., Goradia, D., & Walker, D. (2022). Collaboration structures in integrated healthcare delivery systems: An exploratory study of accountable care organizations. *Manufacturing & Service Operations Management*, 24(3), 1796–1820.
- Levi, R., Paulson, E., & Perakis, G. (2019). Optimal interventions for increasing healthy food consumption among low income households. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3486292
- Lowrey, J., Maestas, D., Beaulieu, T., Headings, A., & Hyder, A. (2023). Peer reviewed: Geospatial determinants of food pantry access in the Mid-Ohio Farmacy Program. *Preventing Chronic Disease*, 20, 20.
- Lucas, J., Batt, R. J., & Soremekun, O. A. (2014). Setting wait times to achieve targeted left-without-being-seen rates. *The American Journal of Emergency Medicine*, 32(4), 342–345.
- Marpadga, S., Fernandez, A., Leung, J., Tang, A., Seligman, H., & Murphy, E. J. (2019). Challenges and successes with food resource referrals for food-insecure patients with diabetes. *The Permanente Journal*, 23.
- Mid-Ohio Food Collective. (2022). Impact Report. https://issuu.com/mofcollective/docs/2022_impact_issu
- Moran, A. J., & Roberto, C. A. (2023). A “Food Is Medicine” approach to disease prevention: Limitations and alternatives. *Journal of the American Medical Association*, 330(23), 2243–2244.
- Morse, S. (2019). SDOH: Food insecurity adds \$53 billion annually to healthcare costs. <https://www.healthcarefinancenews.com/news/sdoh-food-insecurity-adds-53-billion-annually-healthcare-costs>
- Mozaffarian, D., Fleischhacker, S., & Andrés, J. R. (2021). Prioritizing nutrition security in the US. *JAMA*, 325(16), 1605–1606.
- Palakshappa, D., Garg, A., Peltz, A., Wong, C. A., Cholera, R., & Berkowitz, S. A. (2023). Food insecurity was associated with greater family health care expenditures in the US, 2016–17: Study examines the relationship between food insecurity and family health care expenditures. *Health Affairs*, 42(1), 44–52.
- Park, A. (2019). *Why food could Be the best medicine of all*. Time Magazine. <https://time.com/longform/food-best-medicine/>
- Ramdas, K., & Darzi, A. (2017). Adopting innovations in care delivery—the case of shared medical appointments. *New England Journal of Medicine*, 376, 1105–1107.
- Ramdas, K., Teisberg, E. O., & Tucker, A. (2012). Four ways to reinvent service delivery. *Harvard Business Review*, December.
- Ranjit, N., Aiyer, J. N., Touns, J. D., Liew, E., Way, K., Brown, H. S., McWhorter, J. W., & Sharma, S. V. (2023). Clinical outcomes of a large-scale, partnership-based regional food prescription program: Results of a quasi-experimental study. *BMC Research Notes*, 16(1), 1–9.
- Rittenhouse, D. R., Shortell, S. M., & Fisher, E. S. (2009). Primary care and accountable care—Two essential elements of delivery-system reform. *New England Journal of Medicine*, 361(24), 2301–2303.
- Roels, G., & Staats, B. R. (2021). Introduction to the Manufacturing & Service Operations Management Special Issue on People-Centric Operations. *Manufacturing & Service Operations Management*, 23(4), 758–760.
- Salles, A., Awad, M., Goldin, L., Krus, K., et al. (2019). Estimating implicit and explicit gender bias among health care professionals and surgeons. *JAMA Network Open*, 2(7), e196545.
- Seligman, H. K., Bolger, A. F., Guzman, D., López, A., & Bibbins-Domingo, K. (2014). Exhaustion of food budgets at month's end and hospital admissions for hypoglycemia. *Health Affairs*, 33(1), 116–123.
- Seligman, H. K., Levi, R., Adebisi, V. O., Coleman-Jensen, A., Guthrie, J. F., & Frongillo, E. A. (2023). Assessing and monitoring nutrition security to promote healthy dietary intake and outcomes in the United States. *Annual Review of Nutrition*, 43, 409–429.

- Seligman, H. K., & Schillinger, D. (2010). Hunger and socioeconomic disparities in chronic disease. *New England Journal of Medicine*, 363(1), 6–9.
- Senot, C., Chandrasekaran, A., & Ward, P. T. (2016). Collaboration between service professionals during the delivery of health care: Evidence from a multiple-case study in US hospitals. *Journal of Operations Management*, 42, 62–79.
- Simchi-Levi, D. (2020). From the Editor: Diversity, equity, and inclusion in management science. *Management Science*, 66(9), 3802.
- Song, H., Tucker, A. L., & Murrell, K. L. (2015). The diseconomies of queue pooling: An empirical investigation of emergency department length of stay. *Management Science*, 61(12), 3032–3053.
- Song, H., & Veeraraghavan, S. (2018). Quality of care. *Handbook of healthcare analytics: Theoretical Minimum for conducting 21st century Research on healthcare operations*, 79–108.
- Staats, B. R., Dai, H., Hofmann, D., & Milkman, K. L. (2017). Motivating process compliance through individual electronic monitoring: An empirical examination of hand hygiene in healthcare. *Management Science*, 63(5), 1563–1585.
- Stauffer, J. M., Vanajakumari, M., Kumar, S., & Mangapora, T. (2022). Achieving equitable food security: How can food bank mobile pantries fill this humanitarian need. *Production and Operations Management*, 31(4), 1802–1821.
- Stratton, I. M., Adler, A. I., Neil, H. A., Matthews, D. R., et al. (2000). Association of glycaemia with macrovascular and microvascular complications of type 2 diabetes (UKPDS 35): Prospective observational study. *BMJ*, 321, 405–412.
- Sunar, N., & Swaminathan, J. M. (2022). Socially relevant and inclusive operations management. *Production and Operations Management*, 31(12), 4379–4392.
- Swartz, H. (2018). Produce Rx programs for diet-based chronic disease prevention. *AMA Journal of Ethics*, 20(10), 960–973.
- Taylor, L. A., Tan, A. X., Coyle, C. E., Ndumele, C., et al. (2016). Leveraging the social determinants of health: What works? *PLoS One*, 11(8), e0160217.
- Thompson, S., Whitaker, J., Kohli, R., & Jones, C. (2019). Chronic disease management: How IT and analytics create healthcare value through the temporal displacement of care. In S. Thompson, J. Whitaker, R. Kohli, & C. Jones (Eds.), *Chronic disease management: How IT and analytics create healthcare value through the temporal displacement of care*. MIS Quarterly.
- Walker, D. M., Hefner, J. L., MJ, D. P., Garner, J. A., Headings, A., Joseph, J. J., & Clark, A. (2022). Approaches for overcoming barriers to cross-sector data sharing. *American Journal of Managed Care*, 28(1), 11–16.
- Wang, G., Li, J., Hopp, W. J., Fazzalari, F. L., & Bolling, S. F. (2019). Using patient-specific quality information to unlock hidden healthcare capabilities. *Manufacturing & Service Operations Management*, 21(3), 582–601.
- Ward, Z. J., Bleich, S. N., Craddock, A. L., Barrett, J. L., Giles, C. M., Flax, C., Long, M. W., & Gortmaker, S. L. (2019). Projected US state-level prevalence of adult obesity and severe obesity. *New England Journal of Medicine*, 381(25), 2440–2450.
- Xie, L., O'Connor, J., Albert, S., Gary-Webb, T., et al. (2024). Churn in supplemental nutrition assistance program: Changes in Medicaid Expenditure and Acute Care Utilization. *Medical Care*, 62(1), 3–10.
- Zenk, S. N., Tabak, L. A., & Pérez-Stable, E. J. (2022). Research opportunities to address nutrition insecurity and disparities. *JAMA*, 327(20), 1953–1954.

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APPENDIX A: ELECTRONIC COMPANION

A.1 | Sample size limitations

Sample size limitations associated with the unbalanced panel data are of critical importance and ultimately restrict the final analysis. The first three limitations enumerated below are first-order considerations that can be addressed using simple data reduction steps. We address limitation #4 in detail in Appendix Section A.1.1. We cannot address limitation #5 due to the limited availability of data and observational nature of our study design.

1. **FQHC clinic visit frequency.** The frequency of patient visits to the eight FQHC clinics was quite low. All patients were low-income and subject to SDOH that may have influenced the frequency of clinic visits. From June 24, 2015 to May 11, 2018, the FQHC clinics saw 22,930 patients across 97,000 healthcare encounters. This amounts to 1.4 visits per patient per year, on average. When we sample the 3709 food insecure patients referred to Pharmacy, 3053 had fewer than three clinic visits before and three clinic visits after referral. Figure A1 provides sample sizes following each data reduction step.
2. **Prevalence of health outcome data.** Health outcomes were recorded when a patient visited an FQHC

clinic. Health outcomes like BMI (A1c) are more (less) available because patient intake procedures vary. Weight and height are more consistently recorded, whereas an A1c test result must be ordered by a physician.

3. **Low food pantry utilization.** 92% of food insecure patients referred to the Pharmacy program made fewer than 11 food pantry visits ($n = 3415$). Only 7.9% ($n = 294$) had 11 or more visits. The distribution of the number of pantry visits 1 year after referral presents a strong positive skew (see Figure A2).
4. **Lack of clinical data structured around referral.** A pathological comparison might result from unbalanced panel data. This might occur, for example, when a treated participant with clinic visits dispersed around the referral date is compared to a control non-participant with clinic visits more significantly clustered (see Figure A3). Variability in clinic visits (and health outcomes) around referral at $t = 0$ is a form of sample-induced endogeneity (Certo et al. 2016) insofar that the significant clustering of clinic visits in the control group might not adequately capture the change in health quality in the post period. If this were the case, the DiD procedure would not estimate the change in health quality in the control group correctly. Treated and control patients should have

Step	Patient Sample	N	%
Total Sample	1) Total patients	22,930	
Referral Status	2A) Not referred (Food Secure)	19,221	
	2B) Referred (Food Insecure)	3,709	16.2% of 1)
Compliance Status (e.g., Food Pantry Use)	3A) Referred, 0 food pantry visit	2,327	
	3B) Referred, ≥ 1 food pantry visit (<i>Complier</i>)	1,382	37.3% of 2B)
	3C) Referred, ≥ 11 food pantry visits (<i>HighPantryUse</i>)	294	7.9% of 2B)
Prevalence of Health Outcome (BMI)	4A) Referred, ≥ 1 food pantry visit, with BMI before and after referral	842	60.9% of 3B)
	4B) Referred, ≥ 1 food pantry visit, ≥ 3 BMI before and ≥ 3 BMI after referral	364	26.3% of 3B)
	4C) Referred, ≥ 1 food pantry visit, ≥ 3 BMI before and ≥ 3 BMI after referral within [-12,+12] months around referral	265	19.2% of 3B)
Prevalence of Health Outcome (BMI), <i>HighComorbid</i>	6) Referred, ≥ 1 food pantry visit, ≥ 3 BMI before and ≥ 3 BMI after referral within [-12,+12] months around referral Charlson Comorbidity Index ≥ 3	47	17.8% of 4C)
Prevalence of Health Outcome (BMI), <i>HighComorbid</i> , high pantry use (≥ 11)	7) Referred, ≥ 1 food pantry visit, ≥ 3 BMI before and ≥ 3 BMI after referral within [-12,+12] months around referral Charlson Comorbidity Index ≥ 3 ≥ 11 food pantry visits	12	4.5% of 4C)

FIGURE A1 Data reduction steps to address first-order data limitations.

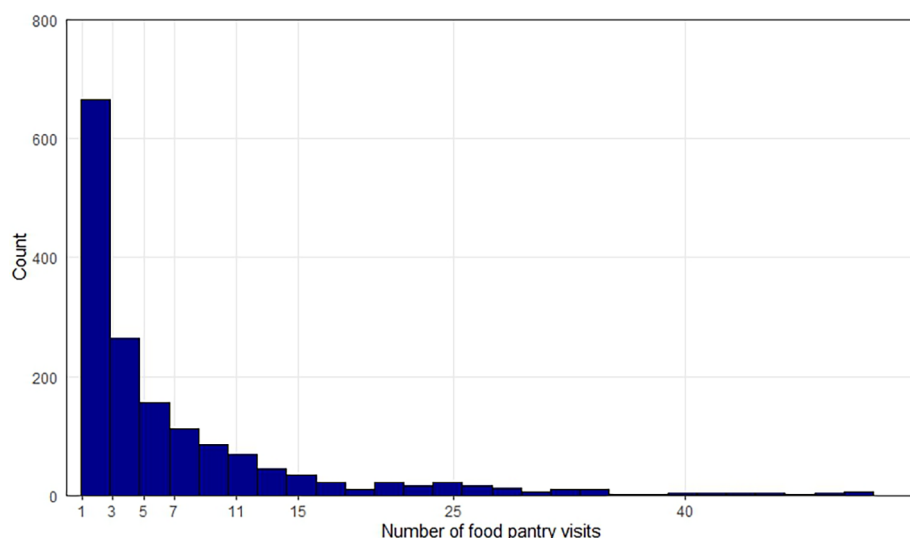


FIGURE A2 Histogram for the number of food pantry visits 1-year after referral for compliers.

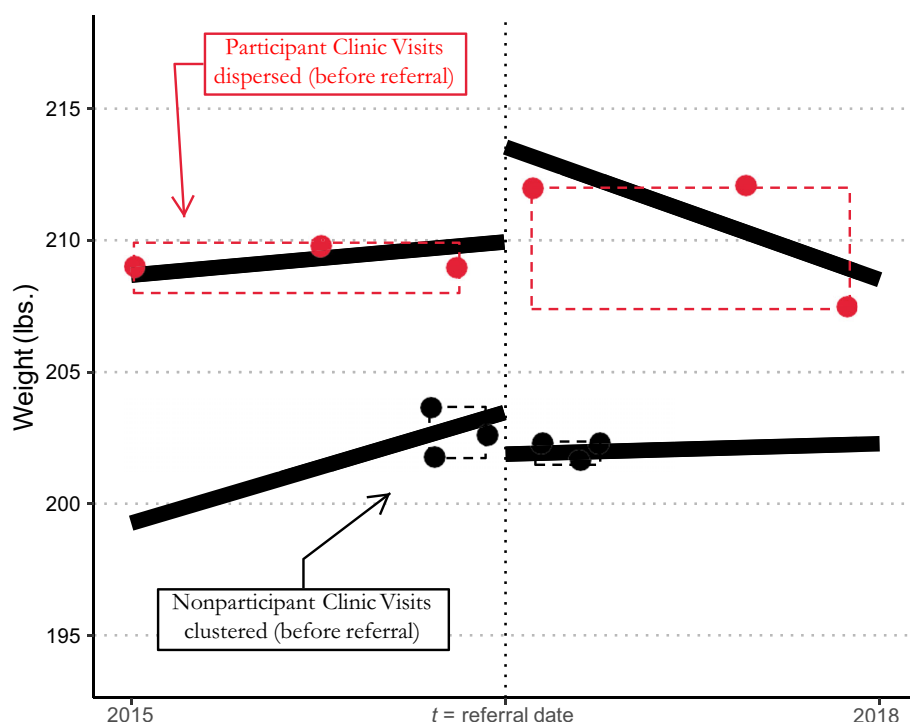


FIGURE A3 The pathological case: comparing health outcome data in an unbalanced panel.

similar longitudinal availability. We use the number of days elapsed to identify a proper control group using unbalanced panel data. Days elapsed has been used by others (Ahuja et al., 2022) and is the number of days between treatment and clinic visit, which essentially normalizes patient clinic visits to a standard time interval. This procedure is briefly discussed and indicated in the footnotes of Table 2.

5. **Lack of clinic-community coordination.** Ideally, patients with high pantry use made their pantry visits immediately after referral and in a routine manner at

regular intervals. To assess a change in health outcomes, the same patients would also have health outcomes measured at a similar interval. However, SDOH-related factors that are the likely culprit of low food pantry use also affect return clinic visit frequency, reducing the coordinated availability of the two types of data.

To study the dose–response relationship and properly evaluate the heterogeneous effect of food pantry use in the Pharmacy program, all five limitations need to be addressed. A randomized control trial is one reasonable approach.

A.1.1. | Addressing lack of clinical data structured around referral

We normalize all clinic visits to a given patient's date of referral. $Days_elapsed_{it}$ captures the number of days between a clinic visit at t and referral at $t=0$. This measure is used in Ahuja et al. (2022) to capture the number of days between consecutive visits. In our study, a negative value for $Days_elapsed_{it}$ indicates a clinic visit occurred before referral; positive values indicate visits after referral. We consider only clinic visits within a 24-month window around referral or $Days_elapsed \in [-365, +365]$.

For the treated sample, we summarize the availability of health outcome data using the mean and standard deviation (SD) of $Days_elapsed$. The treated sample had a mean of -10 and a SD of 210 . A negative mean indicates that the center of the distribution of normalized clinic visits is 10 days before referral. Thus, treated patients had large negative values for $Days_elapsed$ before referral, weighted by small positive values after referral. Therefore, clinic visits were more dispersed before referral and clustered together (and closer in time to referral) after referral. To select control patients, we rely on a propensity score matching method and select patients on the basis on observable demographic characteristics and the mean and SD of $Days_elapsed$. Of the 2257 control patients, 312 have similar demographic characteristics and availability of health outcome data relative to referral, as measured by $Days_elapsed$.

A.2 | Aggregating data to the patient-quarter level of analysis

Several recent studies examine patient health outcome data at the patient-quarter level of analysis (Ahuja et al., 2020; Ahuja et al., 2022). Aggregating patient-visit data at a higher level of analysis facilitates imputation of missing data.⁸ With a more complete panel, we can run additional analyses and test the common assumptions of DiD.

We do this by first aggregating data every quarter and follow the procedure for normalizing clinic visits to the referral date. To normalize visits, we convert patient clinic visits to the corresponding quarter in the calendar year (e.g., 2015Q3: 07/01/2015–09/30/2015) and take the difference in number of quarters between a given clinic visit and the quarter of the referral date. If the referral occurred in quarter $t = 0$, and then $Qtr_diff \in [-4, +4]$.

For this new sample, we return to all 2239 patients with at least one clinic visit before and after referral. Patient-quarter observations totaled 13,008. We impute missing values using the next observation carried backward (NOCB) before referral and last observation carried

forward (LOCF) after referral. LOCF has been used in other studies (Ahuja et al., 2020). Multiple values observed in a given quarter were averaged. This procedure increased the sample size to 20,099 patient quarters.

A.3 | LATE model using 2SLS for endogenous compliance

There remains the possibility that our results are being driven by the endogeneity of compliance. For example, if compliers had slightly higher incomes relative to non-compliers, the former group may be more likely to commit to long-term behavior change and sustain new behaviors like visiting the food pantry. Note, however, that this is unlikely since 100% of the FQHC patient population was at or below 200% of the FPL, which indicates nearly universal resource constraints.

Following Gubler et al. (2018), we rely on the two-stage least-squares (2SLS) method to address endogenous compliance and compare the local average treatment effect (LATE) to our main results. The LATE is obtained by averaging the treatment effect across all compliers, after instrumenting for endogenous compliance. The advantages of the LATE model are reviewed in Choi (2021) and Angrist et al. (1996, p. 467).

Because our DiD model exploits variation within patients to identify a treatment effect, we require an instrument that also varies within-patient. The selected instrument is $Post_{it} \times RefEndMonth_i$, where $RefEndMonth_i$ takes a value of 1 for a patient that received a referral on the 15th day of the month or later and 0 otherwise. Byrne and Just (2021) demonstrate that the demand for food pantry services markedly increases at the end of the month, which suggests that social needs are greater then. Patients referred at the end of the month, therefore, should be more likely to comply with the referral and make at least one visit to the food pantry. The instrument, $Post_{it} \times RefEndMonth_i$, takes a value of 1 for all clinic visits after referral for a patient that was referred at the end of the month, and a 0 otherwise.

This selected instrument satisfies the relevance and validity criteria. To demonstrate relevance, the first-stage instrumental variables regression controlling for patient fixed effects produces a first-stage F statistic of 22.59, which is not weak in the sense of Andrews et al. (2019). Our instrument is also valid as it directly satisfies the exclusion restriction: a patient referred at the end of the month does not affect health outcomes in any reasonable way, other than through their participation in the Pharmacy program. Based on an exploratory data analysis, there are no observable characteristics that predict a referral at the end of the month. The second-stage equation is as follows.

$$\begin{aligned}
 \text{LOG}(\text{Avg_BMI})_{it} = & \alpha_i + \beta_1 \text{Post}_{it} + \beta_2 \text{Post}_{it} \\
 & \times \text{NonComplier}_i + \beta_3 \text{Post}_{it} \times \widehat{\text{Complier}}_i + \beta_4 \text{Post}_{it} \times \widehat{\text{Complier}}_i \times \text{HighComorbid}_i \\
 & + \beta_5 \text{Post}_{it} \times \widehat{\text{Complier}}_i \times \text{HighPantryUse}_i + \beta_6 \text{Post}_{it} \times \widehat{\text{Complier}}_i \times \text{HighComorbid}_i \times \text{HighPantryUse}_i \\
 & + r_{it} + \epsilon_{it}
 \end{aligned}
 \tag{A1}$$

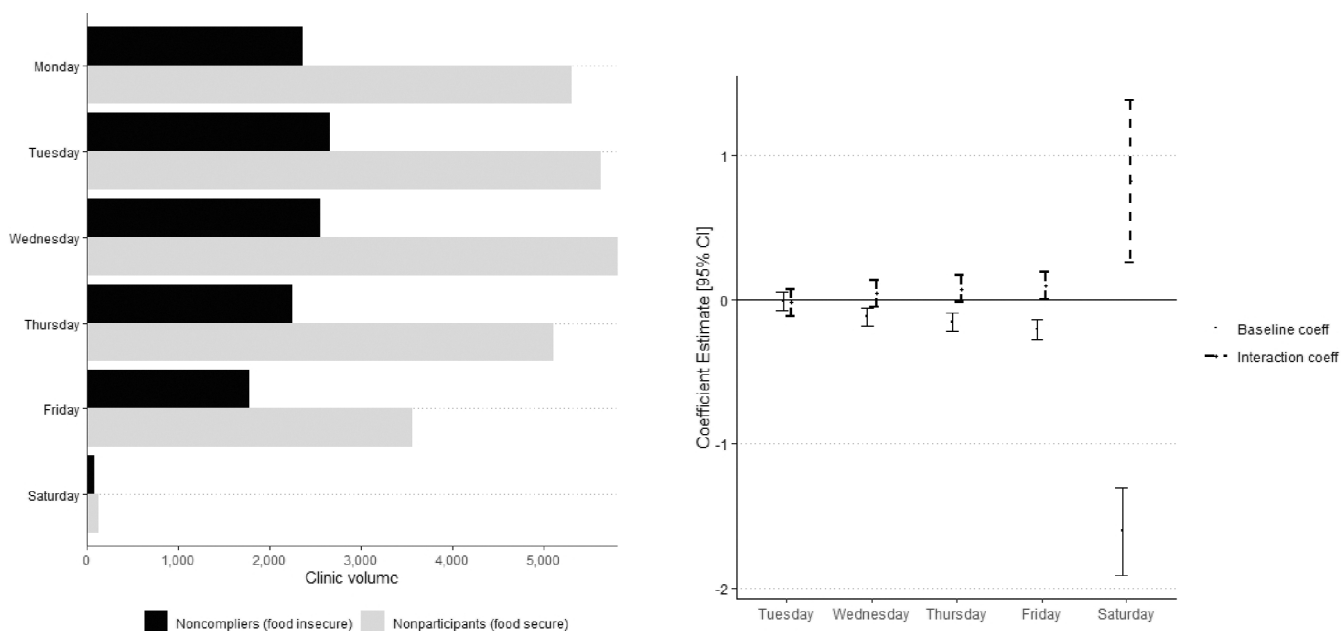


FIGURE A4 Total clinic volume (L) and marginal effect estimates [95%CI] from Equation (3) in Section 6 (R).

where Avg_BMI_{it} is the average Body Mass Index (kg/m^2) for patient i in quarter t . $\text{Post}_{it} \times \widehat{\text{Complier}}_i$ is the expected value of the interaction term, after corrected for the residuals from the first-stage regression, represented by r_{it} near the end of the specification. α_i , ϵ_i , Post_{it} , Noncomplier_i , $\widehat{\text{Complier}}_i$, HighPantryUse_i and HighComorbid_i all maintain their original respective definitions.

Using the patient-quarter level of analysis, the final sample included 1179 patients and 6713 patient-quarters. After adjusting for the possible endogeneity of compliance, we find a non-significant effect of Pharmacy compliance on average BMI for compliers ($\beta = 0.014$, $p = 0.725$) and a marginally significant effect for noncompliers ($\beta = 0.004$, $p = 0.091$). High comorbid and high pantry use compliers had a statistically significant reduction in BMI ($\beta = -0.0350$, $p = 0.00$). LOCF and NOCB were used to impute missing average BMI values and construct a more balanced panel. With imputation, the

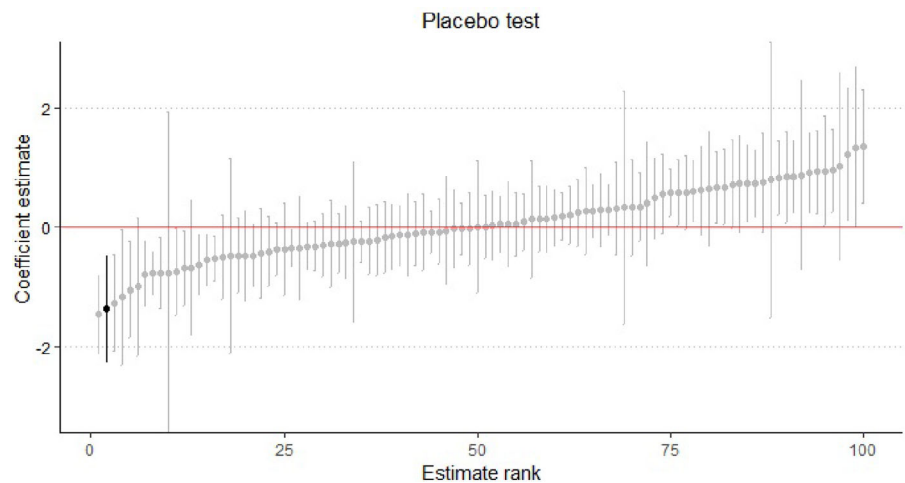
balanced sample included 9-quarter observations per patient or 10,611 total patient-quarters observations. The LATE estimates were robust to this alternative construction of the sample.

A.4 | Clinic visit volume by day-of-week and food security status

In this section, we explore differences in FQHC clinic utilization in terms of total clinic encounters. In acute care, utilization can be used as an indicator of health quality, such as the total number of Emergency Department visits in Ahuja et al. (2020). In primary care, utilization is equally important but may not necessarily be a direct proxy for health quality. It is more likely that primary care utilization corresponds to the patients' social dimensions of care.

Figure A4 provides an overview of total clinic visits by day of the week across all eight FQHCs. We also plot

FIGURE A5 The coefficient estimate and 95% confidence interval for β_6 from 100 falsification tests.



the baseline and interaction coefficients for Day_{jt} from Equation (3). The results reported in Table 3 demonstrate that clinics do not exhibit any significant differences in visit volume on Tuesdays. As the week progresses, however, total clinic volume decreases.

There is statistically significant difference in clinic visit volume by food security status at the end of the week. The positive and significant interaction coefficient for Friday and Saturday in model (3) suggests higher odds of food insecure patient volume on these days. This may be because food insecure patients must resolve competing demands (Seligman & Schillinger, 2010), which partly contributes to a delay in receiving care. Food insecure patients may have to contend with greater occupational demands earlier in the week and postpone routine or primary care appointment to later in the week, such as Friday or Saturday when their occupational demands are likely lower.

Our results show that food insecure patients are more likely to visit the clinic beginning on Friday. The difference in visit probabilities by food security status is most pronounced on Saturday, which, relative to Monday, has a 66% higher likelihood of clinic visit volume being composed of food insecure patients.⁹

A.5 | Placebo test for β_6 in Equation (2)

The placebo test involves several steps. Treated and control groups are first randomly assigned and coefficient estimates obtained using the DiD specification in Equation (2). This procedure is repeated 100 times for 100 falsification tests. Figure A5 presents the resulting coefficients in rank order from smallest to largest.

Model (4) of Table 2 shows β_6 to be significant at the 5% level. This means that we would expect 5 of 100 estimates to be as extreme or more extreme than the observed coefficient. That 99 of the 100 coefficients were larger than

the observed coefficient ($\beta_6 = -1.49$) helps to verify the significance of the observed coefficient and ensures that the results are not due to random assignment of the high comorbid and high pantry use complier group.

A.6 | Relative time models for parallel trends assumption

The relative time model, by examining whether the treatment effect coincides with the treatment date, helps to ensure that the DiD effect estimate is not biased by any time trend in the pre-treatment period. If a pre-treatment trend were to confound the DiD effect, there would be a significant difference in the outcome variable across the treated and control groups in the pre-treatment period.

Conceptually, pre-treatment trends bias the effect estimate if individuals could anticipate a treatment and, prior to receiving it, change their behavior to affect the outcome of interest. In our context, a patient cannot anticipate the date of referral. The clinic visit at which the provider originates the referral to the Farmacy program is like any other clinic visit. Moreover, the preventative health outcome of interest is BMI, which is an indicator of health that does not change instantaneously and thus cannot easily be altered by patients in anticipation of treatment.

Relative time models help to establish the validity of the DiD method and have been used to satisfy the parallel trend assumption. The regression specification is similar to Equation (1), but with a series of time period dummies and their interaction with the treatment group. The omitted time period is the treatment date, against which each time interaction is tested. Ideally, this procedure is repeated for each treated group. We present results only for the complier group, as the other subgroups do not have adequate data. Formally, the specification is,

$$\begin{aligned}
\text{Avg_BMI}_t = & \beta_0 + \beta_1 \text{Qtr}_{t-4} \times \text{Complier} + \beta_2 \text{Qtr}_{t-3} \times \text{Complier} \\
& + \beta_3 \text{Qtr}_{t-2} \times \text{Complier} + \beta_4 \text{Qtr}_{t-1} \times \text{Complier} \\
& + \beta_5 \text{Qtr}_{t+1} \times \text{Complier} + \beta_6 \text{Qtr}_{t+2} \times \text{Complier} \\
& + \beta_7 \text{Qtr}_{t+3} \times \text{Complier} + \beta_8 \text{Qtr}_{t+4} \times \text{Complier} + \epsilon_t
\end{aligned}
\quad (\text{A2})$$

where *Qtr* are the time period dummy variables. Table A1 reveals the coefficients to not be significant in the pre-treatment period, indicating no pre-treatment difference between the treatment and control groups. In the three quarters immediately after treatment, however, treated compliers experienced a reduction in BMI relative to control and noncomplier patients. This is evidence that the treatment effect coincided with the treatment date.

Interestingly, the effect tapers off and becomes non-significant four quarters after treatment. There may be several reasons for this, the first being a simple lack of available data. If patients do not return to the clinic four quarters after treatment, we cannot observe their health outcome for that time period. Second, the non-significance of the coefficient could be an artifact of the LOCF data imputation strategy employed in the post-treatment period. LOCF fills in missing values from the previous quarter. Missing values four quarters from the treatment date can be imputed with data from one, two, or three quarters after treatment. In our low-income patient cohort, the availability of health outcome data decreases with increasing quarters after treatment owing to significant barriers to clinic access and other SDOH that confront these patients.

A.7 | Determinants of food pantry access

Using food pantry data for all compliers available in *PantryTrak*, we summarize key measures of food pantry access at the patient level. Considering the patient characteristics including location and demographic factors

TABLE A1 Relative time models using imputed values and patient-quarter level of analysis.

	Avg_BMI (1)
$\text{Qtr}_{t-4} \times \text{Complier}$	0.11 (0.18)
$\text{Qtr}_{t-3} \times \text{Complier}$	0.08 (0.17)
$\text{Qtr}_{t-2} \times \text{Complier}$	-0.02 (0.17)
$\text{Qtr}_{t-1} \times \text{Complier}$	0.05 (0.16)
$\text{Referral}_t = \text{BASE}$	
$\text{Qtr}_{t+1} \times \text{Complier}$	-0.68*** (0.25)
$\text{Qtr}_{t+2} \times \text{Complier}$	-0.68*** (0.26)
$\text{Qtr}_{t+3} \times \text{Complier}$	-0.49* (0.27)
$\text{Qtr}_{t+4} \times \text{Complier}$	-0.43 (0.28)
Constant	27.17*** (0.56)
Patient FEs	✓
Number patients	1052
Number patient-quarters	9468
Adjusted R^2	0.94
F Statistic	138.20*** (df = 2256; 17,761)

Note: * $p < .1$; ** $p < .05$; *** $p < .01$.

strongly correlated with total visits ($r = 0.64$, $p < .001$). We explore what drives the number of unique locations using the following OLS model,

$$\begin{aligned}
\text{LOG}(\text{NumLocations})_i = & \beta_0 + \beta_1 \text{TotVis}_i \\
& + \beta_2 \text{NumDirections}_i + \underbrace{\beta_3 \text{MeanDistGC}_i + \beta_4 \text{ADI}_i + \beta_5 \text{TransitScore}_i}_{\text{Location factors}} \\
& + \underbrace{\beta_6 \text{Sex}_i + \beta_7 \text{AgeRange}_i + \beta_8 \text{Ethnicity}_i + \beta_9 \text{Race}_i}_{\text{Demographic factors}} + \epsilon_i
\end{aligned}
\quad (\text{A3})$$

and their association with food pantry access, we find the number of unique food pantry locations visited to be

where *NumLocations* is the total number of unique food pantry locations visited, which has a range of [1,16] and

a median of 2, *TotVis* is the total number of food pantry visits, *NumDirections* is the total number of cardinal directions traveled, with a range of [1–4]. The location factors include, *MeanDistGC*, *ADI*, and *TransitScore*. *MeanDistGC* is the mean great-circle distance from the patient's home address to all 330 food pantry locations. A lower number indicates a location with greater pantry density. *ADI* is the area-deprivation index defined at the US Census block group of the patients randomized home address. *TransitScore* is a proxy for the mobility of a particular location and was assessed at the patients' randomized coordinate home location.

Figure A6 reports the results. Positive and significant coefficients for *TotVis* and *NumDirections* demonstrate face validity, as a higher number of total visits and greater number of cardinal directions traveled from a patient's home address should be closely correlated with the number of unique locations. Negative and significant coefficients observed for location factors *MeanDistGC* and *TransitScore* indicate that patients who lived in a location with greater pantry density and lower mobility had higher numbers of unique visits. Lastly, fewer visits to unique locations appear to be made by males than by females. Demographic factors included in Equation (A3) were not significant.

	<i>LOG(NumLocations)</i>
	(1)
<i>TotVis</i>	0.03*** (0.00)
<i>NumDirections</i>	0.52*** (0.03)
<i>MeanDistGC</i>	−0.00*** (0.00)
<i>TransitScore</i>	−0.37** (0.13)
<i>Male</i>	−0.07* (0.03)
<i>ADI</i>	✓
<i>AgeRange</i>	✓
<i>Ethnicity</i>	✓
<i>Race</i>	✓
Number of patients	1,114
Adjusted R ²	0.56
F Statistic	72.1*** (df = 20; 1093)

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

FIGURE A6 OLS results for the total number of unique food pantry locations visited.

A.8 | Nutritional content of food at the pantry

To address the possibility that compliers received different foods or foods of different nutritional quality,¹⁰ we requested food pantry service data from the partnering food bank. Food pantry services were classified as general pantry services (e.g., shelf-stable foods), produce services (e.g., fresh fruits and vegetables), and other services (e.g., non-food items). Due to the small sample sizes and presence of non-normal distributions, we assess variation across the two groups using the non-parametric Kruskal-Wallis H -test.

Among high pantry use compliers, high and low comorbid patients did not differ significantly in total access to food pantry services (e.g., general + produce + other) ($H = 0.08$, $p = 0.77$). Regarding individual services, neither did we find any significant differences in the values of average general pantry ($H = 0.16$, $p = 0.69$), produce ($H = 0.67$, $p = 0.41$) or other ($H = 0.49$, $p = 0.49$).

A.9 | Longitudinal food pantry use among high pantry use compliers

We examine in this section the search, frequency, and consistency of pantry visits for compliers with *high pantry use* (i.e., ≥ 11 food pantry visits in the 1-year period after referral). Separating the high food pantry use complier group by comorbid status to explore whether low comorbid (require less complex care) and high comorbid (require more complex care) patients demonstrate any differences on these three dimensions of food pantry access,¹¹ we find high comorbid patients to be more likely to use food as medicine and thus potentially benefit more from the Pharmacy program.

A.9.1. | Food pantry search behaviors

We examine food pantry search behaviors relative to patients' home addresses and visits to the eight FQHC clinics (clinic visits could reveal additional information about a patient's willingness to travel). We map the geographic location of a given complier's home address and the set of food pantries visited during the 1-year period after referral. We capture driving distance (in miles) and duration (in minutes) for a trip that originates from the complier's home address and ends at a food pantry. Because a patient may make multiple or repeat trips to the same pantry, we also capture the number of trips completed.

Overall, we found high comorbid patients to be willing to visit a greater number of, and travel further to

access, food pantries. Among high pantry use compliers, low (high) comorbid patients visited 4.5 (5.5) unique food pantries and logged an average round trip distance of 6.8 (53.4) miles.¹² Visual inspection of patients' pantry and clinic access location data revealed high comorbid patients to not only visit pantries further from their home addresses, but also along a broader search perimeter. These patients searched for pantries in all four cardinal directions from their home addresses. High and low comorbid patients being similar in SES, a greater search perimeter implies that high comorbid patients were more creative in finding ways to minimize the transaction cost of visits. Interestingly, low comorbid patients visited pantries closer to home and typically located close to the referring FQHC clinic.

One example that demonstrates differences in geographic food pantry search behavior by comorbid status is provided in Figure A7. On the left, the low comorbid complier visited three food pantries, all of which were located to the East of their home address and in the same direction as the FQHC. Based on cartographic features of the study area, the three food pantries were located along a similar route and involved using the same highway as that used to make a trip to either of the FQHC. On the right, however, the high comorbid complier visited two pantries in the same vicinity and directionality as the FQHC (e.g., South-Southeast), but also visited pantries in three other coordinate directions from their home address. The pantry to the West of the patient's home address was visited the most (e.g., $\times 6$ or six times) and had the longest driving distance and duration of 8.3 miles and 12 min, respectively. The other pantries had 1–2 visits each. The high frequency of visits and high driving distance in a direction opposite of the FQHC

suggest that the high comorbid complier developed a creative and sustainable solution for making pantry visits.

Food pantry volunteers indicated that some of the Farmacy patients had greater motivation and ability to visit the food pantry. One volunteer specifically mentioned that some would “piggyback” their food pantry visits off of other planned trips and, for example, swing by the food pantry “on their way home from work” or stop by on the weekend after running errands. Weekend trips seemed to be particularly popular for Farmacy compliers who were also parents and had to arrange for child care or borrow a friend's car to make a visit to the food pantry. Our conversations with volunteers suggested that high comorbid compliers were more likely to employ such solutions.

A.9.2. | Food pantry frequency

We explore differences in frequency by examining the *timing* of the first visit, which is important because it is correlated with the total number of visits. The sooner patients are exposed to a food pantry the more likely they are to make return visits.

We consider the number of days between the referral and the first food pantry visit for each patient. The distribution by comorbid status is provided in Figure A8. On average, low comorbid compliers (top) make their first visit **32 days** after referral, high comorbid compliers (bottom) after only **23 days**. This **12 day** difference suggests that high comorbid compliers may place greater value on, and consequently be more willing to accept the inconvenience of visiting, a food pantry. Food pantry volunteers with whom we spoke referred to the “social stigma” of the

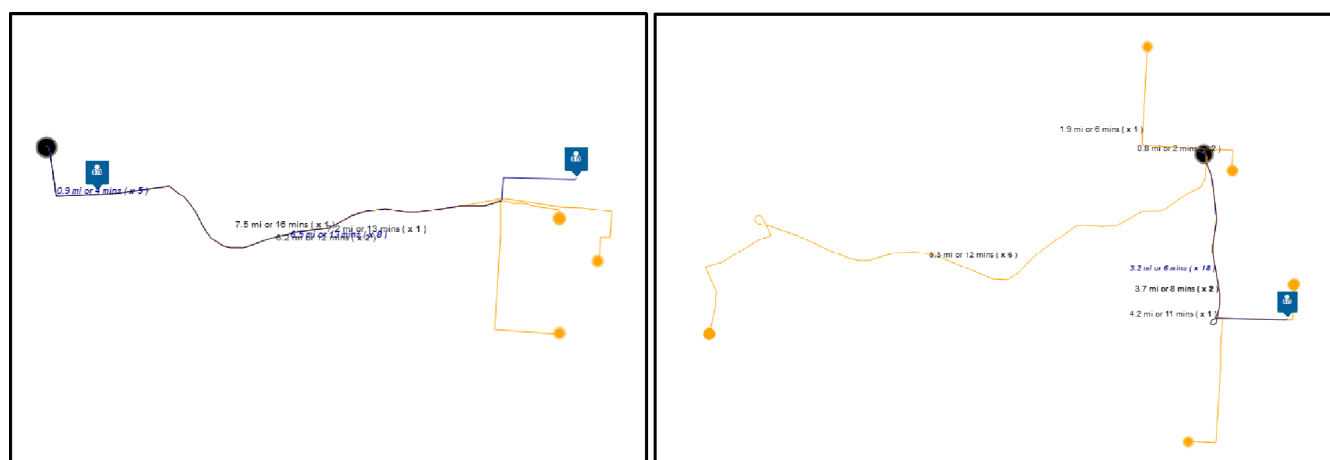


FIGURE A7 The geographic search behaviors for a low (L) and high comorbid (R) complier with high food pantry use (≥ 11 food pantry visits).

visit as one possible explanation. Our finding that high comorbid compliers visited more quickly after referral suggests that they are less influenced by this effect.¹³

A.9.3. | Food pantry consistency

We examine differences in food pantry consistency across the two groups by considering the cumulative number of food pantry visits in each consecutive month after referral. Figure A9 plots the cumulative number of food pantry visits for low (left) and high (right) comorbid patients during the 1-year period after referral. Colors indicate a unique patient; the height of the bar corresponds to the

number of food pantry visits in that month. Behavior appears to vary across the two groups.

Cumulative food pantry visits 0–12 months after referral decrease for low comorbid patients (left), and for high comorbid patients (right) increase from months 0–5, peaking at month 5 and remaining high and relatively stable thereafter. Whereas pantry visits made by high comorbid patients increase with the number of months after referral, greater attrition is observed among low comorbid patients. The cumulative number of food pantry visits is also slightly higher and more stable for the high comorbid group.

Aggregate differences in food pantry consistency by comorbid status are also confirmed by a closer look at

FIGURE A8 Differences in the timing of the first food pantry use 1-year after referral for low (top) and high (bottom) comorbid and high pantry use compliers (≥ 11 food pantry visits).

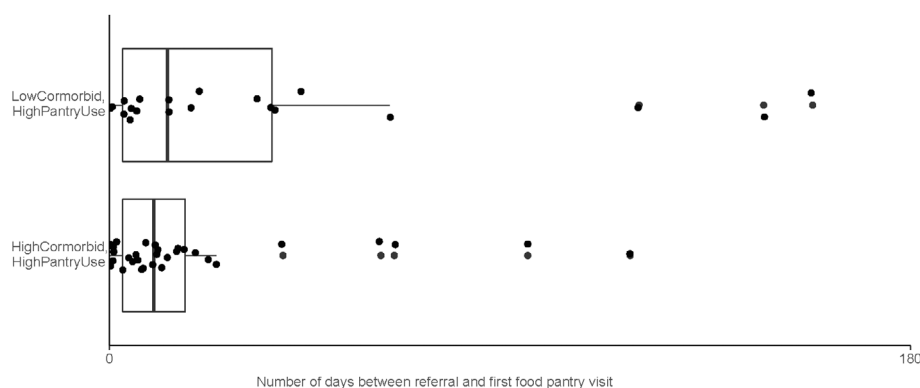
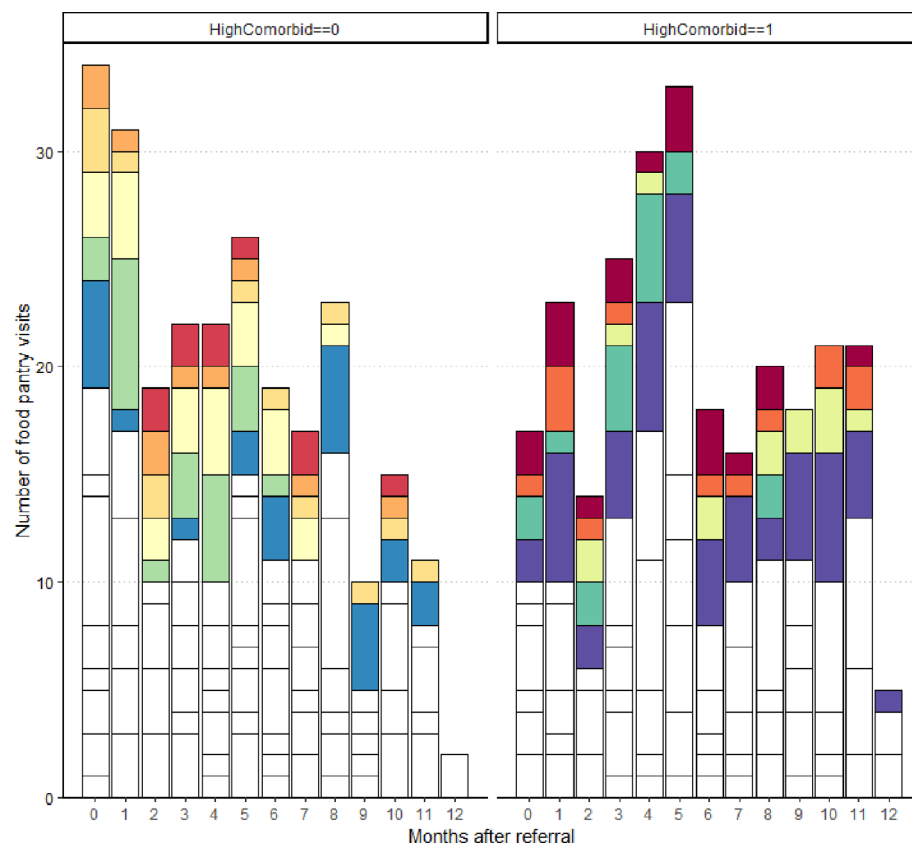


FIGURE A9 Differences in cumulative food pantry visits for low (left) and high comorbid (right) compliers with high food pantry use (≥ 11 food pantry visits). In each chart, a color represents a unique patient and the height of the bar corresponds to their aggregate pantry visits.



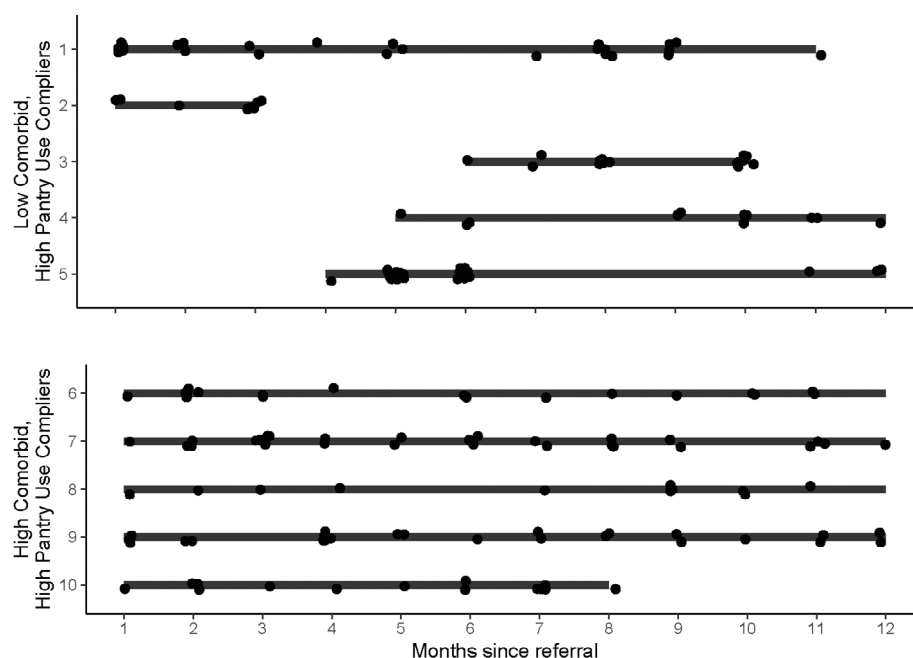


FIGURE A10 Food pantry visits clustered by number of months after referral for low (top) and high (bottom) comorbid compliers with high food pantry use (≥ 11 food pantry visits).

patient-specific differences. We visually inspect individual pantry access patterns using plots like the one in Figure A10, which shows 10 representative high pantry use compliers by comorbid status. We find that high comorbid compliers utilized the pantry more consistently, spreading out visits almost evenly over the 1-year period after referral, whereas visits by low comorbid compliers were more irregular.

Observing pantry access patterns for patients 9 and 3, the former's pantry visits appear to be evenly spaced across all 12 months, whereas the latter's, made during a continuous four-month period from 6 to 10 months after referral, are mostly clumped together in months six and eight. Individual patient-specific food pantry access patterns verify the trends observed in the aggregate data.

A.10 | Qualitative exploration of food pantry use
How does the food received from the food pantry get utilized? Our empirical analyses show that regularly consuming food received from food pantries yields better health outcomes. It is important, however, to understand who consumes the food, which may not be the patient referred to the Farmacy program who may pick up the food and distribute it among other members of the household. Standard practice for all food pantries is to adjust the volume of food distributed to clients' family size. Usually, the total basket of food received at a pantry visit is the equivalent of a two-week supply for client and family. As it concerned the allocation of food, the director of the Farmacy program shared with us the following comment.

A patient visiting a typical pantry would receive a weekly pick-up consisting of 20 to 30 pounds of produce that varies based on what is seasonally available. Fresh whole-grain bread is also typically provided for the patient and household members. All household members should consume this food, as it potentially leads to an overall healthier household.

Another standard practice is to distribute pamphlets or informational brochures on how to prepare, cook, and consume food received from a pantry. It is not unusual for a client visiting a food pantry to see an avocado for the first time. Pantry volunteers hand out pamphlets and interact with clients to ensure that they know what they're taking home and how to consume it. According to the Farmacy program director,

The relationship between food and health is woven throughout the collaborations with our food pantry partners. Our full-time dietitian on-staff works with agencies to provide relevant training on healthful choices and also helps connect local nutrition education resources with agencies through cooperative extensions. Other community groups work with our Mid-Ohio Farmacy partner agencies to promote healthful choices.

Patient education regarding food choices is accomplished through several venues. The dietitian at the food

bank meets several times a year with all food pantry partners to answer questions about the Farmacy program as well as direct pantries to reliable resources for acquainting patients with healthful eating habits (e.g., MyPlate, SNAP-Ed, Celebrate Your PlateA SNAP-Ed program). Among the resources typically available to patients on the Farmacy website is an online extension of the program with recipes and other relevant health information on metabolic conditions. To supplement these resources, the dietitian helps to establish cooperative extensions with nutrition resources available through local health systems. In some cases, partnering healthcare systems' trained, on-staff dietitians have worked directly with Farmacy clients to guide healthy food choices.

Do patients have access to other healthy food options? Common variants of this question include, (1) Are compliers restricted to visiting certain pantry locations and, if yes, how do pantry characteristics (e.g., operating hours, accessibility, food types, etc.) influence the compliance outcome? and (2) Why does a patient who can purchase healthy foods from nearby stores need to visit a food pantry? We discussed these questions with our co-author, a PhD-trained nutritionist who directed the second-phase implementation of the Farmacy program. She first explained the revised rules implemented for the Farmacy program regarding food pantry catchment areas.

Usually, food pantries restrict their services to clients that reside in an immediate geography like a zip code. This is because of the number of pantries that work with us [at Mid-Ohio Food Collective] and additional requirements from their local donors... However, we reached out to our more capable pantries, or a large majority of them, and explained the Farmacy program and the Phase II implementation. We required the pantries to provide food to anyone that walked in the door... because we didn't want a referred patient to be turned away at the pantry.

Our interviews with other food bank leaders and pantry volunteers suggested that a vast majority of patients nevertheless visit food pantries geographically near their homes. Having abundant resources with which to procure food from food banks and other channels, all pantries were well stocked with healthy produce and other perishable items. On-shelf, all pantries thus appeared similar and well stocked across all food categories. Pantry-specific food availability did not affect patient access to fresh food.

To answer the second question, we queried three primary care clinicians at the FQHC about their patients' general diet and food access behaviors. Absent the Farmacy program, we discovered that most patients that presented to the FQHC did not have reliable access to fresh food. To make this point, the lead dietitian at the FQHC told us that,

Without this program, my patients cannot afford to eat healthy. All of them struggle to make ends meet. Talking to an obese patient, for example, I can't say 'If you start eating healthy, you will improve your weight.' They just can't. Many of them don't have the money... or the access to healthy foods. They live in food deserts. How can they afford fresh?

By definition, the FQHC provides care to low-income patients who are uninsured or under-insured. In our study sample, 80% (99%) of patients were below 100% (200%) of the FPL; gross income for a four-person family was thus at or below \$30,000. That financial pressures cause compensatory food consumption behaviors that exacerbate disease has been discussed extensively in the literature (Seligman & Schillinger, 2010).

Are patients improving their health through other means? Although our empirical analyses show the health outcomes of high comorbid compliers who visited the food pantry to have improved, the possibility that these same patients have greater access to other health improvement interventions (e.g., using the gym or health club) cannot be entirely ruled out. However, field interviews with clients and feedback from the FQHC's lead dietitian lead us to believe that participation in health-promoting interventions is a near financial impossibility for the patients in our sample.

Most of the food insecure clients who visit food pantries do not own homes and often rent in unfavorable locations that are generally not in the catchment area for health-promoting community amenities like public or private gyms.

Are high comorbid compliers improving their health as a matter of course? This question poses the possibility that our econometric results are simply a reversion to the mean: high comorbid compliers improve their health because it is easier for them to reduce BMI, and low comorbid compliers should increase BMI. We address this possibility using both quantitative and qualitative evidence. To understand the mechanics of the health quality benefit that attends food pantry access, we interviewed several clients, some in our study and others not, across various pantry locations and created a

storyboard of their experiences that attempted to capture how continued access to fresh food (at food pantries) changed behaviors. In the words of one client,

When we started having access to fresh food, it changed the way my family ate. Instead of stopping by a McDonalds to feed all of us, we... go home and cook food and it would last us a couple of days. Eating healthy

helped everyone [in the family] feel a little better I think... I started coming to the pantry two years ago and I've lost close to 50 lbs. It has changed my life.

Other clients expressed similar sentiments regarding how access to fresh food changed their and their families' eating behaviors, further strengthening our conclusion that food pantry access improves health outcomes.